

CHAPTER

3

Quadratic Equation and Inequalities (Inequalities)

Section-A

JEE Advanced/ IIT-JEE

A Fill in the Blanks

- The coefficient of x^{99} in the polynomial $(x-1)(x-2)\dots(x-100)$ is (1982 - 2 Marks)
- If $2 + i\sqrt{3}$ is a root of the equation $x^2 + px + q = 0$, where p and q are real, then $(p, q) = (\dots, \dots)$. (1982 - 2 Marks)
- If the product of the roots of the equation $x^2 - 3kx + 2e^{2\ln k} - 1 = 0$ is 7, then the roots are real for $k = \dots$. (1984 - 2 Marks)
- If the quadratic equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ ($a \neq b$) have a common root, then the numerical value of $a + b$ is (1986 - 2 Marks)
- The solution of equation $\log_7 \log_5 (\sqrt{x+5} + \sqrt{x}) = 0$ is (1986 - 2 Marks)
- If $x < 0, y < 0, x + y + \frac{x}{y} = \frac{1}{2}$ and $(x + y) \frac{x}{y} = -\frac{1}{2}$, then $x = \dots$ and $y = \dots$. (1990 - 2 Marks)
- Let n and k be positive such that $n \geq \frac{k(k+1)}{2}$. The number of solutions $(x_1, x_2, \dots, x_k), x_1 \geq 1, x_2 \geq 2, \dots, x_k \geq k$, all integers, satisfying $x_1 + x_2 + \dots + x_k = n$, is (1996 - 2 Marks)
- The sum of all the real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is (1997 - 2 Marks)
- If $n_1, n_2, \dots, n_p$ are p positive integers, whose sum is an even number, then the number of odd integers among them is odd. (1985 - 1 Mark)
- If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, where $ac \neq 0$, then $P(x)Q(x) = 0$ has at least two real roots. (1985 - 1 Mark)
- If x and y are positive real numbers and m, n are any positive integers, then $\frac{x^n y^m}{(1+x^{2n})(1+y^{2m})} > \frac{1}{4}$ (1989 - 1 Mark)

C MCQs with One Correct Answer

- If ℓ, m, n are real, $\ell \neq m$, then the roots by the equation: $(\ell - m)x^2 - 5(\ell + m)x - 2(\ell - m) = 0$ are (1979)
 - Real and equal
 - Complex
 - Real and unequal
 - None of these.
- The equation $x + 2y + 2z = 1$ and $2x + 4y + 4z = 9$ have (1979)
 - Only one solution
 - Only two solutions
 - Infinite number of solutions
 - None of these.
- If x, y and z are real and different and (1979)
 $u = x^2 + 4y^2 + 9z^2 - 6yz - 3zx - 2xy$, then u is always.
 - non negative
 - zero
 - non positive
 - none of these
- Let $a > 0, b > 0$ and $c > 0$. Then the roots of the equation $ax^2 + bx + c = 0$ (1979)
 - are real and negative
 - have negative real parts
 - both (a) and (b)
 - none of these
- Both the roots of the equation $(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$ are always (1980)
 - positive
 - real
 - negative
 - none of these.
- The least value of the expression $2 \log_{10} x - \log_x(0.01)$, for $x > 1$, is (1980)
 - 10
 - 2
 - 0.01
 - none of these.
- If $(x^2 + px + 1)$ is a factor of $(ax^3 + bx + c)$, then (1980)
 - $a^2 + c^2 = -ab$
 - $a^2 - c^2 = -ab$
 - $a^2 - c^2 = ab$
 - none of these

B True / False

- For every integer $n > 1$, the inequality $(n!)^{1/n} < \frac{n+1}{2}$ holds. (1981 - 2 Marks)
- The equation $2x^2 + 3x + 1 = 0$ has an irrational root. (1983 - 1 Mark)
- If $a < b < c < d$, then the roots of the equation $(x-a)(x-c) + 2(x-b)(x-d) = 0$ are real and distinct. (1984 - 1 Mark)

8. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is (1982 - 2 Marks)
 (a) 4 (b) 1 (c) 3 (d) 2
9. Two towns A and B are 60 km apart. A school is to be built to serve 150 students in town A and 50 students in town B. If the total distance to be travelled by all 200 students is to be as small as possible, then the school should be built at (1982 - 2 Marks)
 (a) town B (b) 45 km from town A
 (c) town A (d) 45 km from town B
10. If p, q, r are any real numbers, then (1982 - 2 Marks)
 (a) $\max(p, q) < \max(p, q, r)$
 (b) $\min(p, q) = \frac{1}{2}(p + q - |p - q|)$
 (c) $\max(p, q) < \min(p, q, r)$
 (d) none of these
11. The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is (1982 - 2 Marks)
 (a) $-4 < x \leq 0$ (b) $0 < x < 1$
 (c) $-100 < x < 100$ (d) $-\infty < x < \infty$
12. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has (1984 - 2 Marks)
 (a) no root (b) one root
 (c) two equal roots (d) infinitely many roots
13. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in the interval (1984 - 2 Marks)
 (a) $[\frac{1}{2}, 2]$ (b) $[-1, 2]$ (c) $[-\frac{1}{2}, 1]$ (d) $[-1, \frac{1}{2}]$
14. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval - (1985 - 2 Marks)
 (a) $(2, \infty)$ (b) $(1, 2)$
 (c) $(-2, -1)$ (d) none of these
- 15*. If α and β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always (1989 - 2 Marks)
 (a) two real roots
 (b) two positive roots
 (c) two negative roots
 (d) one positive and one negative root
- * Question has more than one correct option.
16. Let a, b, c be real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies (1989 - 2 Marks)
 (a) $\gamma = \frac{\alpha + \beta}{2}$ (b) $\gamma = \alpha + \frac{\beta}{2}$
 (c) $\gamma = \alpha$ (d) $\alpha < \gamma < \beta$
17. The number of solutions of the equation $\sin(e)^x = 5^x + 5^{-x}$ is
 (a) 0 (b) 1 (1990 - 2 Marks)
 (c) 2 (d) Infinitely many
18. Let α, β be the roots of the equation $(x-a)(x-b) = c, c \neq 0$. Then the roots of the equation $(x-\alpha)(x-\beta) + c = 0$ are (1992 - 2 Marks)
 (a) a, c (b) b, c
 (c) a, b (d) $a+c, b+c$
19. The number of points of intersection of two curves $y = 2 \sin x$ and $y = 5x^2 + 2x + 3$ is (1994)
 (a) 0 (b) 1 (c) 2 (d) ∞
20. If p, q, r are +ve and are in A.P., the roots of quadratic equation $px^2 + qx + r = 0$ are all real for (1994)
 (a) $\left| \frac{r}{p} - 7 \right| \geq 4\sqrt{3}$ (b) $\left| \frac{p}{r} - 7 \right| \geq 4\sqrt{3}$
 (c) all p and r (d) no p and r
21. Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is (1994)
 (a) 15 (b) 9 (c) 7 (d) 8
22. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then (1999 - 2 Marks)
 (a) $a < 2$ (b) $2 \leq a \leq 3$
 (c) $3 < a \leq 4$ (d) $a > 4$
23. If α and β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then (2000S)
 (a) $0 < \alpha < \beta$ (b) $\alpha < 0 < \beta < |\alpha|$
 (c) $\alpha < \beta < 0$ (d) $\alpha < 0 < |\alpha| < \beta$
24. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a+b)(c+d)$ satisfies the relation
 (a) $0 \leq M \leq 1$ (b) $1 \leq M \leq 2$ (2000S)
 (c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$
25. If $b > a$, then the equation $(x-a)(x-b) - 1 = 0$ has (2000S)
 (a) both roots in (a, b)
 (b) both roots in $(-\infty, a)$
 (c) both roots in $(b, +\infty)$
 (d) one root in $(-\infty, a)$ and the other in $(b, +\infty)$
26. For the equation $3x^2 + px + 3 = 0, p > 0$, if one of the root is square of the other, then p is equal to (2000S)
 (a) $1/3$ (b) 1 (c) 3 (d) $2/3$
27. If $a_1, a_2, \dots, a_n$ are positive real numbers whose product is a fixed number c , then the minimum value of $a_1 + a_2 + \dots + a_{n-1} + 2a_n$ is (2002S)
 (a) $n(2c)^{1/n}$ (b) $(n+1)c^{1/n}$
 (c) $2nc^{1/n}$ (d) $(n+1)(2c)^{1/n}$
28. The set of all real numbers x for which $x^2 - |x+2| + x > 0$, is (2002S)
 (a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$
29. If $\alpha \in \left(0, \frac{\pi}{2}\right)$ then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is always greater than or equal to (2003S)
 (a) $2 \tan \alpha$ (b) 1 (c) 2 (d) $\sec^2 \alpha$

Quadratic Equation and Inequalities (Inequalities)

30. For all 'x', $x^2 + 2ax + 10 - 3a > 0$, then the interval in which 'a' lies is (2004S)

(a) $a < -5$ (b) $-5 < a < 2$ (c) $a > 5$ (d) $2 < a < 5$

31. If one root is square of the other root of the equation $x^2 + px + q = 0$, then the relation between p and q is (2004S)

(a) $p^3 - q(3p - 1) + q^2 = 0$ (b) $p^3 - q(3p + 1) + q^2 = 0$
(c) $p^3 + q(3p - 1) + q^2 = 0$ (d) $p^3 + q(3p + 1) + q^2 = 0$

32. Let a, b, c be the sides of a triangle where $a \neq b \neq c$ and $\lambda \in R$. If the roots of the equation

$x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then (2006 - 3M, -1)

(a) $\lambda < \frac{4}{3}$ (b) $\lambda > \frac{5}{3}$
(c) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (d) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

33. Let α, β be the roots of the equation $x^2 - px + r = 0$ and

$\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is (2007 - 3 marks)

(a) $\frac{2}{9}(p - q)(2q - p)$ (b) $\frac{2}{9}(q - p)(2p - q)$
(c) $\frac{2}{9}(q - 2p)(2q - p)$ (d) $\frac{2}{9}(2p - q)(2q - p)$

34. Let p and q be real numbers such that $p \neq 0, p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having

$\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is (2010)

(a) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
(b) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
(c) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
(d) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

35. Let (x_0, y_0) be the solution of the following equations

$$(2x)^{\ell n 2} = (3y)^{\ell n 3}$$

$$3^{\ell n x} = 2^{\ell n y}$$

Then x_0 is (2011)

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) 6

36. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If

$a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is (2011)

(a) 1 (b) 2 (c) 3 (d) 4

37. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0$$

have one root in common is (2011)

(a) $-\sqrt{2}$ (b) $-i\sqrt{3}$ (c) $i\sqrt{5}$ (d) $\sqrt{2}$

38. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has (JEE Adv. 2014)

(a) one purely imaginary root
(b) all real roots
(c) two real and two purely imaginary roots
(d) neither real nor purely imaginary roots

39. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x \sec \alpha + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x \tan \theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals (JEE Adv. 2016)

(a) $2(\sec \theta - \tan \theta)$ (b) $2 \sec \theta$
(c) $-2 \tan \theta$ (d) 0

D MCQs with One or More than One Correct

1. For real x, the function $\frac{(x-a)(x-b)}{x-c}$ will assume all real values provided (1984 - 3 Marks)

(a) $a > b > c$ (b) $a < b < c$
(c) $a > c > b$ (d) $a < c < b$

2. If S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains (1986 - 2 Marks)

(a) $\left(-\infty, -\frac{3}{2}\right)$ (b) $\left(-\frac{3}{2}, -\frac{1}{4}\right)$
(c) $\left(-\frac{1}{4}, \frac{1}{2}\right)$ (d) $\left(\frac{1}{2}, 3\right)$

(e) none of these

3. If a, b and c are distinct positive numbers, then the expression $(b+c-a)(c+a-b)(a+b-c) - abc$ is (1986 - 2 Marks)

(a) positive (b) negative
(c) non-positive (d) non-negative
(e) none of these

4. If a, b, c, d and p are distinct real numbers such that $(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0$ then a, b, c, d (1987 - 2 Marks)

(a) are in A. P. (b) are in G. P.
(c) are in H. P. (d) satisfy $ab = cd$
(e) satisfy none of these

5. The equation $x^{3/4}(\log_2 x)^2 + \log_2 x - 5/4 = \sqrt{2}$ has (1989 - 2 Marks)

(a) at least one real solution
(b) exactly three solutions
(c) exactly one irrational solution
(d) complex roots.

6. The product of n positive numbers is unity. Then their sum is (1991 - 2 Marks)
 (a) a positive integer (b) divisible by n
 (c) equal to $n + \frac{1}{n}$ (d) never less than n
7. Number of divisor of the form $4n + 2$ ($n \geq 0$) of the integer 240 is (1998 - 2 Marks)
 (a) 4 (b) 8 (c) 10 (d) 3
8. If $3^x = 4^{x-1}$, then $x =$ (JEE Adv. 2013)
 (a) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (b) $\frac{2}{2 - \log_2 3}$
 (c) $\frac{1}{1 - \log_4 3}$ (d) $\frac{2\log_2 3}{2\log_2 3 - 1}$
9. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? (JEE Adv. 2015)
 (a) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (b) $\left(-\frac{1}{\sqrt{5}}, 0\right)$
 (c) $\left(0, \frac{1}{\sqrt{5}}\right)$ (d) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$
9. Given $n^4 < 10^n$ for a fixed positive integer $n \geq 2$, prove that $(n+1)^4 < 10^{n+1}$. (1980)
10. Let $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$ (1980)
 Find all the real values of x for which y takes real values.
11. For what values of m , does the system of equations
 $3x + my = m$
 $2x - 5y = 20$
 has solution satisfying the conditions $x > 0, y > 0$. (1980)
12. Find the solution set of the system (1980)
 $x + 2y + z = 1;$
 $2x - 3y - w = 2;$
 $x \geq 0; y \geq 0; z \geq 0; w \geq 0.$
13. Show that the equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has no real solution. (1982 - 2 Marks)
14. mn squares of equal size are arranged to form a rectangle of dimension m by n , where m and n are natural numbers. Two squares will be called 'neighbours' if they have exactly one common side. A natural number is written in each square such that the number written in any square is the arithmetic mean of the numbers written in its neighbouring squares. Show that this is possible only if all the numbers used are equal. (1982 - 5 Marks)
15. If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the n -th power of the other, then show that
 $(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0$ (1983 - 2 Marks)

E Subjective Problems

1. Solve for x : $4^x - 3^{x-\frac{1}{2}} = 3^{x+\frac{1}{2}} - 2^{2x-1}$ (1978)
2. If $(m, n) = \frac{(1-x^m)(1-x^{m-1})\dots(1-x^{m-n+1})}{(1-x)(1-x^2)\dots(1-x^n)}$ (1978)
 where m and n are positive integers ($n \leq m$), show that
 $(m, n+1) = (m-1, n+1) + x^{m-n-1}(m-1, n)$.
3. Solve for x : $\sqrt{x+1} - \sqrt{x-1} = 1$. (1978)
4. Solve the following equation for x : (1978)
 $2 \log_x a + \log_{ax} a + 3 \log_{a^2 x} a = 0, a > 0$
5. Show that the square of $\frac{\sqrt{26-15\sqrt{3}}}{5\sqrt{2}-\sqrt{38+5\sqrt{3}}}$ is a rational number. (1978)
6. Sketch the solution set of the following system of inequalities:
 $x^2 + y^2 - 2x \geq 0; 3x - y - 12 \leq 0; y - x \leq 0; y \geq 0$. (1978)
7. Find all integers x for which (1978)
 $(5x-1) < (x+1)^2 < (7x-3)$.
8. If α, β are the roots of $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s . Deduce the condition that the equations have a common root. (1979)
16. Find all real values of x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \leq 0$ (1983 - 2 Marks)
17. Solve for x : $(5+2\sqrt{6})^{x^2-3} + (5-2\sqrt{6})^{x^2-3} = 10$ (1985 - 5 Marks)
18. For $a \leq 0$, determine all real roots of the equation $x^2 - 2a|x - a| - 3a^2 = 0$ (1986 - 5 Marks)
19. Find the set of all x for which $\frac{2x}{(2x^2+5x+2)} > \frac{1}{(x+1)}$ (1987 - 3 Marks)
20. Solve $|x^2 + 4x + 3| + 2x + 5 = 0$ (1988 - 5 Marks)
21. Let a, b, c be real. If $ax^2 + bx + c = 0$ has two real roots α and β , where $\alpha < -1$ and $\beta > 1$, then show that $1 + \frac{c}{a} + \left|\frac{b}{a}\right| < 0$. (1995 - 5 Marks)
22. Let S be a square of unit area. Consider any quadrilateral which has one vertex on each side of S . If a, b, c , and d denote the lengths of the sides of the quadrilateral, prove that $2 \leq a^2 + b^2 + c^2 + d^2 \leq 4$. (1997 - 5 Marks)

Quadratic Equation and Inequalities (Inequalities)

23. If α, β are the roots of $ax^2 + bx + c = 0$, ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, ($A \neq 0$) for some constant δ , then prove that $\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$.
(2000 - 4 Marks)
24. Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β . (2001 - 4 Marks)
25. If $x^2 + (a - b)x + (1 - a - b) = 0$ where $a, b \in \mathbb{R}$ then find the values of a for which equation has unequal real roots for all values of b . (2003 - 4 Marks)
26. If a, b, c are positive real numbers. Then prove that $(a+1)^7(b+1)^7(c+1)^7 > 7^7 a^4 b^4 c^4$ (2004 - 4 Marks)
27. Let a and b be the roots of the equation $x^2 - 10cx - 11d = 0$ and those of $x^2 - 10ax - 11b = 0$ are c, d then the value of $a + b + c + d$, when $a \neq b \neq c \neq d$, is. (2006 - 6M)

H Assertion & Reason Type Questions

1. Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$ and $\alpha, \frac{1}{\beta}$ are the roots of the equation $ax^2 + 2bx + c = 0$, where $\beta^2 \notin \{-1, 0, 1\}$

STATEMENT - 1: $(p^2 - q)(b^2 - ac) \geq 0$
and

STATEMENT - 2: $b \neq pa$ or $c \neq qa$ (2008)

- (a) STATEMENT - 1 is True, STATEMENT - 2 is True; STATEMENT - 2 is a correct explanation for STATEMENT - 1
(b) STATEMENT - 1 is True, STATEMENT - 2 is True; STATEMENT - 2 is **NOT** a correct explanation for STATEMENT - 1
(c) STATEMENT - 1 is True, STATEMENT - 2 is False
(d) STATEMENT - 1 is False, STATEMENT - 2 is True

I Integer Value Correct Type

1. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations :

$$\begin{aligned} 3x - y - z &= 0 \\ -3x + z &= 0 \\ -3x + 2y + z &= 0 \end{aligned}$$
Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is (2009)
2. The smallest value of k , for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is (2009)
3. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} where $a > 0$ is (2011)
4. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is (2011)

Section-B JEE Main / AIEEE

1. If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having α/β and β/α as its roots is [2002]
 (a) $3x^2 - 19x + 3 = 0$ (b) $3x^2 + 19x - 3 = 0$
 (c) $3x^2 - 19x - 3 = 0$ (d) $x^2 - 5x + 3 = 0$
2. Difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then [2002]
 (a) $a + b + 4 = 0$ (b) $a + b - 4 = 0$
 (c) $a - b - 4 = 0$ (d) $a - b + 4 = 0$
3. Product of real roots of the equation $t^2x^2 + |x| + 9 = 0$ [2002]
 (a) is always positive (b) is always negative
 (c) does not exist (d) none of these
4. If p and q are the roots of the equation $x^2 + px + q = 0$, then [2002]
 (a) $p = 1, q = -2$ (b) $p = 0, q = 1$
 (c) $p = -2, q = 0$ (d) $p = -2, q = 1$
5. If a, b, c are distinct +ve real numbers and $a^2 + b^2 + c^2 = 1$ then $ab + bc + ca$ is [2002]
 (a) less than 1 (b) equal to 1
 (c) greater than 1 (d) any real no.
6. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}, \frac{b}{a}$ and $\frac{c}{b}$ are in [2003]
 (a) Arithmetic - Geometric Progression
 (b) Arithmetic Progression
 (c) Geometric Progression
 (d) Harmonic Progression.
7. The value of ' a ' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is [2003]
 (a) $-\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $-\frac{2}{3}$ (d) $\frac{1}{3}$
8. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$ is
 (a) 3 (b) 2 (c) 4 (d) 1
9. The real number x when added to its inverse gives the minimum value of the sum at x equal to [2003]
 (a) -2 (b) 2 (c) 1 (d) -1
10. Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation [2004]
 (a) $x^2 - 18x - 16 = 0$ (b) $x^2 - 18x + 16 = 0$
 (c) $x^2 + 18x - 16 = 0$ (d) $x^2 + 18x + 16 = 0$
11. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its root are [2004]
 (a) -1, 2 (b) -1, 1 (c) 0, -1 (d) 0, 1

12. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is [2004]
- (a) 4 (b) 12 (c) 3 (d) $\frac{49}{4}$
13. In a triangle PQR , $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ and $-\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0$, $a \neq 0$ then [2005]
- (a) $a = b + c$ (b) $c = a + b$
(c) $b = c$ (d) $b = a + c$
14. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval [2005]
- (a) (5, 6] (b) (6, ∞) (c) $(-\infty, 4)$ (d) [4, 5]
15. If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then the value of $2 + q - p$ is [2006]
- (a) 2 (b) 3 (c) 0 (d) 1
16. All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4, lie in the interval [2006]
- (a) $-2 < m < 0$ (b) $m > 3$
(c) $-1 < m < 3$ (d) $1 < m < 4$
17. If x is real, the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is [2006]
- (a) $\frac{1}{4}$ (b) 41 (c) 1 (d) $\frac{17}{7}$
18. If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is [2007]
- (a) (3, ∞) (b) $(-\infty, -3)$ (c) (-3, 3) (d) (-3, ∞).
19. **Statement-1** : For every natural number $n \geq 2$,

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}.$$

Statement-2 : For every natural number $n \geq 2$,

$$\sqrt{n(n+1)} < n+1.$$
 [2008]
- (a) Statement -1 is false, Statement-2 is true
 (b) Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
 (c) Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
 (d) Statement -1 is true, Statement-2 is false
20. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then the common root is [2009]
- (a) 1 (b) 4 (c) 3 (d) 2
21. If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x, the expression $3b^2x^2 + 6bcx + 2c^2$ is : [2009]
- (a) less than $4ab$ (b) greater than $-4ab$
(c) less than $-4ab$ (d) greater than $4ab$
22. If $\left|z - \frac{4}{z}\right| = 2$, then the maximum value of $|Z|$ is equal to : [2009]
- (a) $\sqrt{5} + 1$ (b) 2 (c) $2 + \sqrt{2}$ (d) $\sqrt{3} + 1$
23. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009} =$ [2010]
- (a) -1 (b) 1 (c) 2 (d) -2
24. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has : [2012]
- (a) infinite number of real roots
(b) no real roots
(c) exactly one real root
(d) exactly four real roots
25. The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in [0, 1] [JEE M 2013]
- (a) lies between 1 and 2 (b) lies between 2 and 3
(c) lies between -1 and 0 (d) does not exist.
26. The number of values of k, for which the system of equations : [JEE M 2013]
- $$(k+1)x + 8y = 4k$$
- $$kx + (k+3)y = 3k - 1$$
- has no solution, is
- (a) infinite (b) 1 (c) 2 (d) 3
27. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is [JEE M 2013]
- (a) 1 : 2 : 3 (b) 3 : 2 : 1 (c) 1 : 3 : 2 (d) 3 : 1 : 2
28. If $a \in \mathbb{R}$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval: [JEE M 2014]
- (a) (-2, -1) (b) $(-\infty, -2) \cup (2, \infty)$
(c) $(-1, 0) \cup (0, 1)$ (d) (1, 2)
29. Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r are in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is: [JEE M 2014]
- (a) $\frac{\sqrt{34}}{9}$ (b) $\frac{2\sqrt{13}}{9}$ (c) $\frac{\sqrt{61}}{9}$ (d) $\frac{2\sqrt{17}}{9}$
30. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE M 2015]
- (a) 3 (b) -3 (c) 6 (d) -6
31. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is : [JEE M 2016]
- (a) 6 (b) 5
(c) 3 (d) -4

3

Quadratic Equation and Inequalities

Section-A : JEE Advanced/ IIT-JEE

- A** 1. -5050 2. -4, 7 3. 2 4. 1 5. 4

6. $-1/4, -1/4$

$$7. \frac{\left[k + \left(n - \frac{k(k+1)}{2} \right) - 1 \right]!}{\left[n - \frac{k(k+1)}{2} \right]! (k-1)!}$$

8. 4

- B** 1. T 2. F 3. T 4. F 5. T 6. F
C 1. (c) 2. (d) 3. (a) 4. (c) 5. (b) 6. (b) 7. (c) 8. (a)
 9. (c) 10. (b) 11. (d) 12. (a) 13. (c) 14. (a) 15. (d) 16. (d)
 17. (a) 18. (c) 19. (a) 20. (b) 21. (c) 22. (a) 23. (b) 24. (a)
 25. (d) 26. (c) 27. (a) 28. (b) 29. (a) 30. (b) 31. (a) 32. (a)
 33. (d) 34. (b) 35. (c) 36. (c) 37. (c) 38. (d) 39. (c)
D 1. (c, d) 2. (a, d) 3. (b) 4. (b) 5. (a, b, c) 6. (d) 7. (a) 8. (a, b, c)
 9. (a, d)

- E** 1. $\frac{3}{2}$ 3. $\frac{5}{4}$ 4. $a^{-1/2}, a^{-4/3}$ 7. 3
 8. $q(r-p)^2 - p(r-p)(s-q) + (s-q)^2$; $(q-s)^2 = (r-p)(ps-qr)$ 10. $[-1, 2) \cup [3, \infty)$
 11. $m \in \left(-\infty, \frac{-15}{2} \right) \cup (30, \infty)$ 12. $x=1, y=0, z=0, w=0$ 16. $[-1, 1) \cup (2, 4]$
 17. $\pm 2, \pm \sqrt{2}$ 18. $\{a \pm a\sqrt{2}, -a \pm a\sqrt{6}\}$ 19. $(-2, -1) \cup \left(-\frac{2}{3}, -\frac{1}{2} \right)$
 20. $-4, -1-\sqrt{3}$ 24. $\alpha^2\beta, \alpha\beta^2$ 25. $a > 1$ 27. 1210

- H** 1. (b)

- I** 1. 7 2. 2 3. 8 4. 2

Section-B : JEE Main/ AIEEE

1. (a) 2. (a) 3. (a) 4. (a) 5. (a) 6. (d) 7. (b) 8. (c)
 9. (c) 10. (b) 11. (c) 12. (d) 13. (b) 14. (c) 15. (b) 16. (c)
 17. (b) 18. (c) 19. (b) 20. (d) 21. (b) 22. (a) 23. (b) 24. (b)
 25. (d) 26. (b) 27. (a) 28. (c) 29. (b) 30. (a) 31. (c)

Section-A JEE Advanced/ IIT-JEE

A. Fill in the Blanks

1. Given polynomial :
 $(x-1)(x-2)(x-3) \dots (x-100)$
 $= x^{100} - (1+2+3+\dots+100)x^{99} + (\dots)x^{98} + \dots$
 Here coeff. of $x^{99} = -(1+2+3+\dots+100)$

$$= \frac{-100 \times 101}{2} = -5050.$$
2. As p and q are real; and one root is $2 + i\sqrt{3}$, other should be $2 - i\sqrt{3}$
 Then $p = -(\text{sum of roots}) = -4$,
 $q = \text{product of roots} = 4 + 3 = 7.$
3. The given equation is $x^2 - 3kx + 2e^{2 \ln k} - 1 = 0$
 Or $x^2 - 3kx + (2k^2 - 1) = 0$
 Here product of roots $= 2k^2 - 1$

$$\therefore 2k^2 - 1 = 7 \Rightarrow k^2 = 4 \Rightarrow k = 2, -2$$

Now for real roots we must have $D \geq 0$

$$\Rightarrow 9k^2 - 4(2k^2 - 1) \geq 0 \Rightarrow k^2 + 4 \geq 0$$

Which is true for all k . Thus $k = 2, -2$

But for $k = -2$, $\ln k$ is not defined

$\therefore$ Rejecting $k = -2$, we get $k = 2$.

4. $\therefore x = 1$ reduces both the equations to $1 + a + b = 0$

$\therefore 1$ is the common root. for $a + b = -1$

$\therefore$ Numerical value of $a + b = 1$

5. $\log_7 \log_5 (\sqrt{x+5} + \sqrt{x}) = 0$

$$\Rightarrow \log_5 (\sqrt{x+5} + \sqrt{x}) = 1 \quad \text{NOTE THIS STEP}$$

$$\Rightarrow \sqrt{x+5} + \sqrt{x} = 5 \Rightarrow x+5 = 25 + x - 10\sqrt{x}$$

$$\Rightarrow 2 = \sqrt{x} \Rightarrow x = 4 \text{ which satisfies the given equation.}$$

6. Given $x < 0, y < 0$

$$x + y + \frac{x}{y} = \frac{1}{2} \text{ and } (x+y) \cdot \frac{x}{y} = -\frac{1}{2}$$

$$\text{Let } x+y = a \text{ and } \frac{x}{y} = b \quad \dots (1)$$

$$\therefore \text{ We get } a+b = \frac{1}{2} \text{ and } ab = -\frac{1}{2}$$

$$\text{Solving these two, we get } a + \left(-\frac{1}{2a}\right) = \frac{1}{2}$$

$$\Rightarrow 2a^2 - a - 1 = 0 \Rightarrow a = 1, -1/2 \Rightarrow b = -1/2, 1$$

$$\therefore (1) \Rightarrow x+y = 1 \text{ and } \frac{x}{y} = -\frac{1}{2}$$

$$\text{or } x+y = \frac{-1}{2} \text{ and } \frac{x}{y} = 1 \text{ But } x, y < 0$$

$$\therefore x+y < 0 \Rightarrow x+y = \frac{-1}{2} \text{ and } \frac{x}{y} = 1$$

On solving, we get $x = -1/4$ and $y = -1/4$.

7. We have

$$x_1 + x_2 + \dots + x_k = n \quad \dots (1)$$

where $x_1 \geq 1, x_2 \geq 2, x_3 \geq 3, \dots, x_k \geq k$; all integers

$$\text{Let } y_1 = x_1 - 1, y_2 = x_2 - 2, \dots, y_k = x_k - k$$

so that $y_1, y_2, \dots, y_k \geq 0$

Substituting the values of $x_1, x_2, \dots, x_k$ in equation .. (1)

$$\text{We get } y_1 + y_2 + \dots + y_k = n - (1 + 2 + 3 + \dots + k)$$

$$= n - \frac{k(k+1)}{2} \quad \dots (2)$$

Now keeping in mind that number of solutions of the equation

$$\alpha + 2\beta + 3\gamma + \dots + q\theta = n$$

for $\alpha, \beta, \gamma, \dots, \theta \in \mathbb{I}$ and each is ≥ 0 , is given by coeff of x^n in

$$(1+x+x^2+\dots)(1+x^2+x^4+\dots)$$

$$(1+x^3+x^6+\dots)\dots(1+x^q+x^{2q}+\dots)$$

We find that no. of solutions of equation (2)

$$= \text{coeff of } x^{n - \frac{k(k+1)}{2}} \text{ in } (1+x+x^2+\dots)^k$$

NOTE THIS STEP

$$= \text{coeff of } x^{n - \frac{k(k+1)}{2}} \text{ in } (1-x)^{-k}$$

$$= \text{coeff of } x^{n - \frac{k(k+1)}{2}} \text{ in } (1 + {}^k C_1 x + {}^{k+1} C_2 x^2$$

$$+ {}^{k+2} C_3 x^3 + \dots) = {}^{k + \left(n - \frac{k(k+1)}{2}\right) - 1} C_{n - \frac{k(k+1)}{2}}$$

$$= \frac{\left[k + \left(n - \frac{k(k+1)}{2}\right) - 1\right]!}{\left[n - \frac{k(k+1)}{2}\right]! (k-1)!}$$

8. $|x-2|^2 + |x-2| - 2 = 0$

Case 1. $x \geq 2$

$$\Rightarrow (x-2)^2 + (x-2) - 2 = 0$$

$$\Rightarrow x^2 - 3x = 0 \Rightarrow x(x-3) = 0$$

$$\Rightarrow x = 0, 3 \text{ (0 is rejected as } x \geq 2)$$

$$\Rightarrow x = 3 \quad \dots (1)$$

Case 2. $x < 2$

$$\{-(x-2)\}^2 - (x-2) - 2 = 0$$

$$\Rightarrow x^2 + 4 - 4x - x = 0 \Rightarrow (x-1)(x-4) = 0$$

$$\Rightarrow x = 1, 4 \text{ (4 is rejected as } x < 2)$$

$$\Rightarrow x = 1 \quad \dots (2)$$

Therefore, the sum of the roots is $3 + 1 = 4$.

B. True/False

1. Consider n numbers, namely $1, 2, 3, 4, \dots, n$.

KEY CONCEPT : Now using A.M. > G.M. for distinct numbers, we get

$$\frac{1+2+3+4+\dots+n}{n} > (1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot n)^{1/n}$$

$$\Rightarrow \frac{n(n+1)}{2n} > (n!)^{1/n} \Rightarrow (n!)^{1/n} < \frac{n+1}{2} \therefore \text{True}$$

2. $2x^2 + 3x + 1 = 0 \Rightarrow x = -1, -1/2$ both are rational

$\therefore$ Statement is FALSE.

Quadratic Equation and Inequalities (Inequalities)

3. $f(x) = (x-a)(x-c) + 2(x-b)(x-d)$.
 $f(a) = +ve; f(b) = -ve; f(c) = -ve; f(d) = +ve$
 $\therefore$ There exists two real and distinct roots one in the interval (a, b) and other in (c, d) . Hence, (True).
4. Consider $N = n_1 + n_2 + n_3 + \dots + n_p$, where N is an even number. Let k numbers among these p numbers be odd, then $p-k$ are even numbers. Now sum of $(p-k)$ even numbers is even and for N to be an even number, sum of k odd numbers must be even which is possible only when k is even.
 $\therefore$ The given statement is false.
5. $P(x) \cdot Q(x) = (ax^2 + bx + c)(-ax^2 + bx + c)$
 $\Rightarrow D_1 = b^2 - 4ac$ and $D_2 = b^2 + 4ac$
 clearly, $D_1 + D_2 = 2b^2 \geq 0$
 $\therefore$ atleast one of D_1 and D_2 is $(+ve)$. Hence, atleast two real roots.
 Thus, (True)
6. As x and y are positive real numbers and m and n are positive integers

$$\therefore \frac{1+x^{2n}}{2} \geq (1 \times x^{2n})^{1/2} \quad \text{and} \quad \frac{1+y^{2m}}{2} \geq (1 \times y^{2m})^{1/2}$$

{For two +ve numbers A.M. $\geq$ G.M.}

$$\Rightarrow \left(\frac{1+x^{2n}}{2} \right) \geq x^n \quad \dots(1)$$

$$\text{and} \left(\frac{1+y^{2m}}{2} \right) \geq y^m \quad \dots(2)$$

Multiplying (1) and (2), we get

$$\frac{(1+x^{2n})(1+y^{2m})}{4} \geq x^n y^m \Rightarrow \frac{1}{4} \geq \frac{x^n y^m}{(1+x^{2n})(1+y^{2m})}$$

Hence the statement is false.

C. MCQs with ONE Correct Answer

1. (c) ℓ, m, n are real, $\ell \neq m$
 Given equation is
 $(\ell - m)x^2 - 5(\ell + m)x - 2(\ell - m) = 0$
 $D = 25(\ell + m)^2 + 8(\ell - m)^2 > 0, \ell, m \in \mathbb{R}$
 $\therefore$ Roots are real and unequal.
2. (d) The given equations are
 $x + 2y + 2z = 1 \quad \dots(1)$
 and $2x + 4y + 4z = 9 \quad \dots(2)$
 Subtracting $(1) \times (2)$ from (2) , we get $0 = 7$ (not possible)
 $\therefore$ No solution.
3. (a) $u = x^2 + 4y^2 + 9z^2 - 6yx - 3zx - 2xy$
 $= \frac{1}{2}[2x^2 + 8y^2 + 18z^2 - 12yz - 6zx - 4xy]$
 $= \frac{1}{2}[(x^2 - 4xy + 4y^2) + (4y^2 + 9z^2 - 12yz)$
 $+ (x^2 + 9z^2 - 6zx)]$

$$= \frac{1}{2}[(x-2y)^2 + (2y-3z)^2 + (3z-x)^2] \geq 0$$

$\therefore u$ is always non-negative.

4. (c) As $a, b, c > 0$, a, b, c should be real (note that order relation is not defined in the set of complex numbers)
 $\therefore$ Roots of equation are either real or complex conjugate. Let α, β be the roots of $ax^2 + bx + c = 0$, then

$$\alpha + \beta = -\frac{b}{a} = -ve, \quad \alpha\beta = \frac{c}{a} = +ve$$

$\Rightarrow$ Either both α, β are $-ve$ (if roots are real) or both α, β have $-ve$ real parts (if roots are complex conjugate)

5. (b) The given equation is

$$(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$$

$$\Rightarrow 3x^2 - 2(a+b+c)x + (ab+bc+ca) = 0$$

$$\text{Discriminant} = 4(a+b+c)^2 - 12(ab+bc+ca)$$

$$= 4[a^2 + b^2 + c^2 - ab - bc - ca]$$

$$= 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0 \quad \forall a, b, c$$

$\therefore$ Roots of given equation are always real.

6. (b) Let $y = 2 \log_{10} x - \log_x 0.01$

$$= 2 \log_{10} x - \frac{\log_{10} 0.01}{\log_{10} x} = 2 \log_{10} x + \frac{2}{\log_{10} x}$$

$$= 2 \left[\log_{10} x + \frac{2}{\log_{10} x} \right]$$

[Here $x > 1 \Rightarrow \log_{10} x > 0$]

Now since sum of a real +ve number and its reciprocal is always greater than or equal to 2.

$\therefore y \geq 2 \times 2 \Rightarrow y \geq 4, \therefore$ Least value of y is 4.

7. (c) As $(x^2 + px + 1)$ is a factor of $ax^3 + bx + c$, we can assume that zeros of $x^2 + px + 1$ are α, β and that of $ax^3 + bx + c$ be α, β, γ so that

$$\alpha + \beta = -p \quad \dots(i)$$

$$\alpha\beta = 1 \quad \dots(ii)$$

$$\text{and } \alpha + \beta + \gamma = 0 \quad \dots(iii)$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{b}{a} \quad \dots(iv)$$

$$\alpha\beta\gamma = \frac{-c}{a} \quad \dots(v)$$

Solving (ii) and (v) we get $\gamma = -c/a$.

Also from (i) and (iii) we get $\gamma = p$

$$\therefore p = \gamma = -c/a$$

Using equations (i), (ii) and (iv) we get

$$1 + \gamma(-p) = \frac{b}{a}$$

$$\Rightarrow 1 + \left(-\frac{c}{a}\right)\left(\frac{c}{a}\right) = \frac{b}{a} \quad (\text{using } \gamma = p = -c/a)$$

$$\Rightarrow 1 - \frac{c^2}{a^2} = \frac{b}{a} \Rightarrow a^2 - c^2 = ab$$

$\therefore$ (c) is the correct answer.

8. (a) $|x|^2 - 3|x| + 2 = 0$
Case I : $x < 0$ then $|x| = -x$
 $\Rightarrow x^2 + 3x + 2 = 0 \Rightarrow (x+1)(x+2) = 0$
 $x = -1, -2$ (both acceptable as < 0)
Case II: $x > 0$ then $|x| = x$
 $\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0$
 $x = 1, 2$ (both acceptable as > 0)
 $\therefore$ There are 4 real solutions.
9. (c) Let the distance of school from A = x
 $\therefore$ The distance of the school from B = $60 - x$
 Total distance covered by 200 students
 $= 2[150x + 50(60 - x)] = 2[100x + 3000]$
 This is min., when $x = 0$
 $\therefore$ school should be built at town A.
10. (b) if $p = 5, q = 3, r = 2$
 $\max(p, q) = 5$; $\max(p, q, r) = 5$
 $\Rightarrow \max(p, q) = \max(p, q, r)$
 $\therefore$ (a) is not true. Similarly we can show that (c) is not true.
- Also $\min(p, q) = \frac{1}{2}(p + q - |p - q|)$
 Let $p < q$ then LHS = p
 and R.H.S. = $\frac{1}{2}(p + q - q + p) = p$
 Similarly, we can prove that (b) is true for $q < p$ too.
11. (d) Given expression $x^{12} - x^9 + x^4 - x + 1 = f(x)$ (say)
 For $x < 0$ put $x = -y$ where $y > 0$
 then we get $f(x) = y^{12} + y^9 + y^4 + y + 1 > 0$ for $y > 0$
 For $0 < x < 1$, $x^9 < x^4 \Rightarrow -x^9 + x^4 > 0$
 Also $1 - x > 0$ and $x^{12} > 0$
 $\Rightarrow x^{12} - x^9 + x^4 + 1 - x > 0 \Rightarrow f(x) > 0$
 For $x > 1$
 $f(x) = x(x^3 - 1)(x^8 + 1) + 1 > 0$
 So $f(x) > 0$ for $-\infty < x < \infty$.
12. (a) Given equation is $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$
 Clearly $x \neq 1$ for the given eq. to be defined. If
 $x - 1 \neq 0$, we can cancel the common term $\frac{-2}{x-1}$ on
 both sides to get $x = 1$, but it is not possible. So given
 eq. has no roots.
 $\therefore$ (a) is the correct answer.
13. (c) Given that $a^2 + b^2 + c^2 = 1$ (1)
 We know $(a + b + c)^2 \geq 0$
 $\Rightarrow a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 0$
 $\Rightarrow 2(ab + bc + ca) \geq -1$ [Using (1)]
 $\Rightarrow ab + bc + ca \geq -1/2$ (2)
 Also we know that

$$\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$$

- $\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca \geq 0$
 $\Rightarrow ab + bc + ca \leq 1$ [Using (1)](2)
 Combining (2) and (3), we get
 $-1/2 \leq ab + bc + ca \leq 1 \therefore ab + bc + ca \in [-1/2, 1]$
 $\therefore$ (c) is the correct answer.
14. (a) First of all for $\log(x-1)$ to be defined, $x-1 > 0$
 $\Rightarrow x > 1$ (1)
 Now, $\log_{0.3}(x-1) < \log_{0.09}(x-1)$
 $\Rightarrow \log_{0.3}(x-1) < \log_{(0.3)^2}(x-1)$
 $\Rightarrow \log_{0.3}(x-1) < \frac{1}{2} \log_{0.3}(x-1)$
 $\Rightarrow 2 \log_{0.3}(x-1) < \log_{0.3}(x-1)$
 $\Rightarrow \log_{0.3}(x-1)^2 < \log_{0.3}(x-1)$
 $\Rightarrow (x-1)^2 > (x-1)$ **NOTE THIS STEP**
 [The inequality is reversed since base lies between 0 and 1]
 $\Rightarrow (x-1)^2 - (x-1) > 0 \Rightarrow (x-1)(x-2) > 0$ (2)
 Combining (1) and (2) we get $x > 2$
 $\therefore x \in (2, \infty)$
15. (d) From the first method,
 $q = \alpha\beta, r = \alpha^4 + \beta^4$ (3)
 Product of the roots of the equation
 $x^2 - 4qx + (2q^2 - r) = 0$
 $= 2q^2 - r = 2\alpha^2\beta^2 - \alpha^4 - \beta^4 = -(\alpha^2 - \beta^2)^2$ [From (3)]
 $= -(\text{positive quantity}) = -\text{ve quantity}$
 $\Rightarrow$ one root is positive and other is negative.
16. (d) **KEY CONCEPT :** If $f(\alpha)$ and $f(\beta)$ are of opposite signs then there must lie a value γ between α and β such that $f(\gamma) = 0$.
 a, b, c are real numbers and $a \neq 0$.
 As α is a root of $a^2x^2 + bx + c = 0$
 $\therefore a^2\alpha^2 + b\alpha + c = 0$ (1)
 Also β is a root of $a^2x^2 - bx - c = 0 \therefore$
 $a^2\beta^2 - b\beta - c = 0$ (2)
 Now, let $f(x) = a^2x^2 + 2bx + 2c$
 Then $f(\alpha) = a^2\alpha^2 + 2b\alpha + 2c = a^2\alpha^2 + 2(b\alpha + c)$
 $= a^2\alpha^2 + 2(-a^2\alpha^2)$ [Using eq. (1)]
 $= -a^2\alpha^2$
 and $f(\beta) = a^2\beta^2 + 2b\beta + 2c$
 $= a^2\beta^2 + 2(b\beta + c)$
 $= a^2\beta^2 + 2(a^2\beta^2)$ [Using eq. (2)]
 $= 3a^2\beta^2 > 0$.
 Since $f(\alpha)$ and $f(\beta)$ are of opposite signs and γ is a root of equation $f(x) = 0$
 $\therefore \gamma$ must lie between α and β
 Thus $\alpha < \gamma < \beta$. $\therefore$ (d) is the correct option.
17. (a) The given eq. is $\sin(e^x) = 5^x + 5^{-x}$
 We know 5^x and 5^{-x} both are +ve real numbers using
 AM $\geq$ GM $\therefore 5^x + \frac{1}{5^x} \geq 2 \Rightarrow 5^x + 5^{-x} \geq 2$

Quadratic Equation and Inequalities (Inequalities)

∴ R.H.S. of given eq. ≥ 2

While $\sin e^x \in [-1, 1]$ i.e. LHS $\in [-1, 1]$

∴ The equation is not possible for any real value of x .
Hence (a) is the correct answer.

18. (c) α, β are roots of the equation $(x-a)(x-b)=c, c \neq 0$

∴ $(x-a)(x-b)-c=(x-\alpha)(x-\beta)$

$\Rightarrow (x-\alpha)(x-\beta)+c=(x-a)(x-b)$

$\Rightarrow$ roots of $(x-\alpha)(x-\beta)+c=0$ are a and b .

∴ (c) is the correct option.

19. (a) We have

$$y = 5x^2 + 2x + 3 = 5\left(x + \frac{1}{5}\right)^2 + \frac{14}{5} > 2, \forall x \in R$$

while $y = 2 \sin x \leq 2, \forall x \in R$

$\Rightarrow$ The two curves do not meet at all.

20. (b) For real roots $q^2 - 4pr \geq 0$

$$\Rightarrow \left(\frac{p+r}{2}\right)^2 - 4pr \geq 0 \quad (\because p, q, r \text{ are in A.P.})$$

$$\Rightarrow p^2 + r^2 - 14pr \geq 0 \Rightarrow \frac{p^2}{r^2} - 14\frac{p}{r} + 1 \geq 0$$

$$\Rightarrow \left(\frac{p}{r} - 7\right)^2 - 48 \geq 0 \Rightarrow \left|\frac{p}{r} - 7\right| \geq 4\sqrt{3}$$

21. (c) For the equation $px^2 + qx + 1 = 0$ to have real roots

$$D \geq 0 \Rightarrow q^2 \geq 4p$$

If $p = 1$ then $q^2 \geq 4 \Rightarrow q = 2, 3, 4$

If $p = 2$ then $q^2 \geq 8 \Rightarrow q = 3, 4$

If $p = 3$ then $q^2 \geq 12 \Rightarrow q = 4$

If $p = 4$ then $q^2 \geq 16 \Rightarrow q = 4$

∴ No. of req. equations = 7.

22. (a) **KEY CONCEPT :** If both roots of a quadratic equation $ax^2 + bx + c = 0$ are less than k then $af(k) > 0, D \geq 0, \alpha + \beta < 2k$.

$$f(x) = x^2 - 2ax + a^2 + a - 3 = 0,$$

$$f(3) > 0, \alpha + \beta < 6, D \geq 0.$$

$$\Rightarrow a^2 - 5a + 6 > 0, a < 3, -4a + 12 \geq 0$$

$$\Rightarrow a < 2 \text{ or } a > 3, a < 3, a < 3 \Rightarrow a < 2.$$

23. (b) Given $c < 0 < b$ and $\alpha + \beta = -b$ (1)

$$\alpha\beta = c \quad \dots(2)$$

From (2), $c < 0 \Rightarrow \alpha\beta < 0 \Rightarrow$ either α is -ve or β is -ve and second quantity is positive.

from (1), $b > 0 \Rightarrow -b < 0 \Rightarrow \alpha + \beta < 0 \Rightarrow$ the sum is negative

$\Rightarrow$ modulus of negative quantity is $>$ modulus of positive quantity but $\alpha < \beta$ is given. Therefore, it is clear that α is negative and β is positive and modulus of α is greater than modulus of $\beta \Rightarrow \alpha < 0 < \beta < |\alpha|$

24. (a) As A.M. $\geq$ G.M. for positive real numbers, we get

$$\frac{(a+b)+(c+d)}{2} \geq \sqrt{(a+b)(c+d)} \Rightarrow M \leq 1$$

(Putting values)

$$\text{Also } (a+b)(c+d) > 0 \quad [\because a, b, c, d > 0]$$

$$\therefore 0 \leq M \leq 1$$

25. (d) The given equation is $(x-a)(x-b)-1=0, b > a$.

$$\text{or } x^2 - (a+b)x + ab - 1 = 0$$

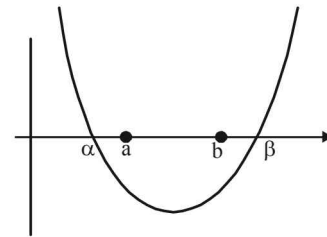
$$\text{Let } f(x) = x^2 - (a+b)x + ab - 1$$

Since coeff. of x^2 i.e. $1 > 0$, $\therefore$ it represents upward parabola, intersecting x -axis at two points. (corresponding to two real roots, D being +ve). Also

$$f(a) = f(b) = -1 \Rightarrow \text{curve is below } x\text{-axis at } a \text{ and } b$$

$\Rightarrow a$ and b both lie between the roots.

Thus the graph of given eqⁿ is as shown.



from graph it is clear that one root of the equation lies in $(-\infty, a)$ and other in (b, ∞) .

26. (c) Let α, α^2 be the roots of $3x^2 + px + 3$.

$$\therefore \alpha + \alpha^2 = -p/3 \text{ and } \alpha^3 = 1$$

$$\Rightarrow (\alpha - 1)(\alpha^2 + \alpha + 1) = 0 \Rightarrow \alpha = 1 \text{ or } \alpha^2 + \alpha = -1$$

If $\alpha = 1, p = -6$ which is not possible as $p > 0$

$$\text{If } \alpha^2 + \alpha = -1 \Rightarrow -p/3 = -1 \Rightarrow p = 3.$$

27. (a) We have

$$\frac{(a_1 + a_2 + \dots + a_{n-1} + 2a_n)}{n} \geq (a_1 a_2 \dots a_{n-1} 2a_n)^{1/n}$$

[Using A.M. $\geq$ G.M.]

$$\Rightarrow a_1 + a_2 + a_3 + \dots + a_{n-1} + 2a_n \geq n(2c)^{1/n}$$

28. (b) For $x < -2$, $|x+2| = -(x+2)$ and the inequality becomes

$$x^2 + x + 2 + x > 0 \Rightarrow (x+1)^2 + 1 > 0$$

which is valid $\forall x \in R$ but $x < -2$

$$\therefore x \in (-\infty, -2) \quad \dots(1)$$

For $x \geq -2$, $|x+2| = x+2$ and the inequality becomes

$$x^2 - x - 2 + x > 0 \Rightarrow x^2 > 2 \Rightarrow x > \sqrt{2} \text{ or } x < -\sqrt{2}$$

$$\text{i.e., } x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

$$\text{but } x \geq -2 \Rightarrow x \in [-2, -\sqrt{2}) \cup (\sqrt{2}, \infty) \quad \dots(2)$$

From (1) and (2)

$$x \in (-\infty, -2) \cup [-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

$$\Rightarrow x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

29. (a) Let $a = \sqrt{x^2 + x}$ and $b = \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$

then using AM $\geq$ GM, we get $\frac{a+b}{2} \geq \sqrt{ab}$

$$\Rightarrow a+b \geq 2\sqrt{ab}$$

$$\Rightarrow \sqrt{x^2+x} + \frac{\tan^2 \alpha}{\sqrt{x^2+x}} \geq 2\sqrt{\tan^2 \alpha} = 2 \tan \alpha$$

$$[\because \alpha \in (0, \pi/2)]$$

30. (b) **KEY CONCEPT :** $f(x) = ax^2 + bx + c$ has same sign as that of a if $D < 0$.

$$x^2 + 2ax + 10 - 3a > 0 \forall x$$

$$\Rightarrow D < 0 \Rightarrow 4a^2 - 4(10 - 3a) < 0 \Rightarrow a^2 + 3a - 10 < 0$$

$$\Rightarrow (a+5)(a-2) < 0 \Rightarrow a \in (-5, 2)$$

31. (a) $x^2 + px + q = 0$

Let roots be α and α^2

$$\Rightarrow \alpha + \alpha^2 = -p, \alpha\alpha^2 = q \Rightarrow \alpha = q^{1/3}$$

$$\therefore (q)^{1/3} + (q^{1/3})^2 = -p$$

Taking cube of both sides, we get

$$q + q^2 + 3q(q^{1/3} + q^{2/3}) = -p^3$$

$$\Rightarrow q + q^2 - 3pq = -p^3 \Rightarrow p^3 + q^2 - q(3p-1) = 0$$

32. (a) $\therefore a, b, c$ are sides of a triangle and $a \neq b \neq c$

$$\therefore |a-b| < |c| \Rightarrow a^2 + b^2 - 2ab < c^2$$

Similarly, we have

$$b^2 + c^2 - 2bc < a^2 ; c^2 + a^2 - 2ca < b^2$$

On adding, we get

$$a^2 + b^2 + c^2 < 2(ab + bc + ca)$$

$$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} < 2 \quad \dots(1)$$

$\therefore$ Roots of the given equation are real

$$\therefore (a+b+c)^2 - 3\lambda(ab+bc+ca) \geq 0$$

$$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} \geq 3\lambda - 2 \quad \dots(2)$$

From (1) and (2), we get $3\lambda - 2 < 2 \Rightarrow \lambda < \frac{4}{3}$.

33. (d) As α, β are the roots of $x^2 - px + r = 0$

$$\therefore \alpha + \beta = p \quad \dots(1)$$

$$\text{and } \alpha\beta = r \quad \dots(2)$$

Also $\frac{\alpha}{2}, 2\beta$ are the roots of $x^2 - qx + r = 0$

$$\therefore \frac{\alpha}{2} + 2\beta = q \text{ or } \alpha + 4\beta = 2q \quad \dots(3)$$

Solving (1) and (3) for α and β , we get

$$\beta = \frac{1}{3}(2q - p) \text{ and } \alpha = \frac{2}{3}(2q - q)$$

Substituting values of α and β , in equation (2),

$$\text{we get } \frac{2}{9}(2p - q)(2q - p) = r.$$

34. (b) Given that $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$

$$\Rightarrow (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = q$$

$$\Rightarrow -p^3 - 3\alpha\beta(-p) = q \Rightarrow \alpha\beta = \frac{p^3 + q}{3p}$$

Now for required quadratic equation,

$$\text{sum of roots} = \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{p^2 - 2\left(\frac{p^3 + q}{3p}\right)}{\frac{p^3 + q}{3p}}$$

$$= \frac{3p^3 - 2p^3 - 2q}{p^3 + q} = \frac{p^3 - 2q}{p^3 + q}$$

$$\text{and product of roots} = \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

$$\therefore \text{Required equation is } x^2 - \left(\frac{p^3 - 2q}{p^3 + q}\right)x + 1 = 0$$

$$\text{or } (p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$$

35. (c) We have $(2x)^{\ln 2} = (3y)^{\ln 3}$

$$\Rightarrow \ln 2. \ln 2x = \ln 3. \ln 3y$$

$$\Rightarrow \ln 2. \ln 2x = \ln 3. (\ln 3 + \ln y) \quad \dots(1)$$

$$\text{Also given } 3^{\ln x} = 2^{\ln y}$$

$$\Rightarrow \ln x. \ln 3 = \ln y. \ln 2 \Rightarrow \ln y = \frac{\ln x. \ln 3}{\ln 2}$$

Substituting this value of $\ln y$ in equation (1), we get

$$\ln 2. \ln 2x = \ln 3 \left[\ln 3 + \frac{\ln x. \ln 3}{\ln 2} \right]$$

$$\Rightarrow (\ln 2)^2 \ln 2x = (\ln 3)^2 \ln 2 + (\ln 3)^2 \ln x$$

$$\Rightarrow (\ln 2)^2 \ln 2x = (\ln 3)^2 (\ln 2 + \ln x)$$

$$\Rightarrow (\ln 2)^2 \ln 2x - (\ln 3)^2 \ln 2x = 0$$

$$\Rightarrow [(\ln 2)^2 - (\ln 3)^2] \ln 2x = 0 \Rightarrow \ln 2x = 0$$

$$\Rightarrow 2x = 1 \text{ or } x = \frac{1}{2}$$

36. (c) $\therefore \alpha, \beta$ are the roots of $x^2 - 6x - 2 = 0$

$$\therefore \alpha^2 - 6\alpha - 2 = 0$$

$$\Rightarrow \alpha^{10} - 6\alpha^9 - 2\alpha^8 = 0$$

$$\Rightarrow \alpha^{10} - 2\alpha^8 = 6\alpha^9 \quad \dots(1)$$

$$\text{Similarly } \beta^{10} - 2\beta^8 = 6\beta^9 \quad \dots(2)$$

From equation (1) and (2)

$$\alpha^{10} - \beta^{10} - 2(\alpha^8 - \beta^8) = 6(\alpha^9 - \beta^9)$$

$$\Rightarrow a_{10} - 2a_8 = 6a_9 \Rightarrow \frac{a_{10} - 2a_8}{2a_9} = 3$$

Quadratic Equation and Inequalities (Inequalities)

37. (b) Let α be the common root of given equations, then

$$\alpha^2 + b\alpha - 1 = 0 \quad \dots(1)$$

$$\text{and } \alpha^2 + \alpha + b = 0 \quad \dots(2)$$

Subtracting (2) from (1), we get

$$(b-1)\alpha - (b+1) = 0$$

$$\text{or } \alpha = \frac{b+1}{b-1}$$

Substituting this value of α in equation (1), we get

$$\left(\frac{b+1}{b-1}\right)^2 + b\left(\frac{b+1}{b-1}\right) - 1 = 0 \text{ or } b^3 + 3b = 0$$

$$\Rightarrow b = 0, i\sqrt{3}, -i\sqrt{3}$$

38. (d) Quadratic equation with real coefficients and purely imaginary roots can be considered as

$$p(x) = x^2 + a = 0 \text{ where } a > 0 \text{ and } a \in R$$

$$\text{The } p[p(x)] = 0 \Rightarrow (x^2 + a)^2 + a = 0$$

$$\Rightarrow x^4 + 2ax^2 + (a^2 + a) = 0$$

$$\Rightarrow x^2 = \frac{-2a \pm \sqrt{4a^2 - 4a^2 - 4a}}{2}$$

$$\Rightarrow x^2 = -a \pm \sqrt{a} i$$

$$\Rightarrow x = \sqrt{-a \pm \sqrt{a} i} = \alpha \pm i\beta \text{ where } \alpha, \beta \neq 0$$

$\therefore p[p(x)] = 0$ has complex roots which are neither purely real nor purely imaginary.

39. (c) $x^2 - 2x \sec \theta + 1 = 0 \Rightarrow x = \sec \theta \pm \tan \theta$
and $x^2 + 2x \tan \theta - 1 = 0 \Rightarrow x = -\tan \theta \pm \sec \theta$

$$\therefore -\frac{\pi}{6} < \theta < -\frac{\pi}{12}$$

$$\Rightarrow \sec \frac{\pi}{6} > \sec \theta > \sec \frac{\pi}{12}$$

$$\text{and } -\tan \frac{\pi}{6} < \tan \theta < -\frac{\tan \pi}{12}$$

$$\text{also } \tan \frac{\pi}{12} < -\tan \theta < \tan \frac{\pi}{6}$$

$$\alpha_1, \beta_1 \text{ are roots of } x^2 - 2x \sec \theta + 1 = 0$$

$$\text{and } \alpha_1 > \beta_1$$

$$\therefore \alpha_1 = \sec \theta - \tan \theta \text{ and } \beta_1 = \sec \theta + \tan \theta$$

$$\alpha_2, \beta_2 \text{ are roots of } x^2 + 2x \tan \theta - 1 = 0 \text{ and } \alpha_2 > \beta_2$$

$$\therefore \alpha_2 = -\tan \theta + \sec \theta, \beta_2 = -\tan \theta - \sec \theta$$

$$\therefore \alpha_1 + \beta_2 = \sec \theta - \tan \theta - \tan \theta - \sec \theta = -2 \tan \theta$$

D. MCQs with ONE or MORE THAN ONE Correct

1. (c,d) Let $y = \frac{(x-a)(x-b)}{(x-c)}$

$$\Rightarrow (x-c)y = x^2 - (a+b)x + ab$$

$$\Rightarrow x^2 - (a+b+y)x + ab + cy = 0$$

$$\text{Here, } \Delta = (a+b+y)^2 - 4(ab+cy)$$

$$= y^2 + 2y(a+b-2c) + (a-b)^2$$

Since x is real and y assumes all real values.

$$\therefore \Delta \geq 0 \text{ for all real values of } y$$

$$\Rightarrow y^2 + 2y(a+b-2c) + (a-b)^2 \geq 0$$

Now we know that the sign of a quad is same as of coeff of y^2 provided its discriminant $B^2 - 4AC < 0$

$$\text{This will be so if, } 4(a+b-2c)^2 - 4(a-b)^2 < 0$$

$$\text{or } 4(a+b-2c+a-b)(a+b-2c-a+b) < 0$$

$$\Rightarrow 16(a-c)(b-c) < 0$$

$$\Rightarrow 16(c-a)(c-b) < 0 \quad \dots(1)$$

Now,

If $a < b$ then from inequation (1), we get $c \in (a, b)$

$$\Rightarrow a < c < b$$

or If $a > b$ then from inequation (1) we get, $c \in (b, a)$

$$\Rightarrow b < c < a \text{ or } a > c > b$$

Thus, we observe that both (c) and (d) are the correct answer.

2. (a,d) **KEY CONCEPT :** Wavy curve method :

$$\text{Let } f(x) = (x-\alpha_1)(x-\alpha_2)\dots(x-\alpha_n)$$

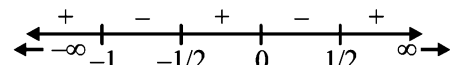
To find sign of $f(x)$, plot $\alpha_1, \alpha_2, \dots, \alpha_n$ on number line in ascending order of magnitude. Starting from right extreme put +ve, -ve signs alternately. $f(x)$ is positive in the intervals having +ve sign and negative in the intervals having -ve sign.

We have,

$$f(x) = \frac{2x-1}{2x^3+3x^2+x} = \frac{2x-1}{x(2x+1)(x+1)}$$

NOTE THIS STEP : Critical points are $x = 1/2, 0, -1/2, -1$

On number line by wavy method, we have



For $f(x) > 0$, when

$$x \in (-\infty, -1) \cup (-1/2, 0) \cup (1/2, \infty)$$

Clearly S contains $(-\infty, -3/2)$ and $(1/2, 3)$

3. (b) Given that a, b, c are distinct +ve numbers. The expression whose sign is to be checked is $(b+c-a)(c+a-b)(a+b-c) - abc$.

As this expression is symmetric in a, b, c , without loss of generality, we can assume that $a < b < c$.

Then $c-a = +ve$ and $c-b = +ve$

$$\therefore b+c-a = +ve \text{ and } c+a-b = +ve$$

But $a+b-c$ may be +ve or -ve.

Case I : If $a+b-c = +ve$ then we can say that a, b, c , are such that sum of any two is greater than the 3rd. Consider

$$x = a+b-c, \quad y = b+c-a, \quad z = c+a-b$$

then x, y, z all are +ve.

$$\text{and then } a = \frac{x+z}{2}, b = \frac{y+x}{2}, c = \frac{z+y}{2}$$

Now we know that A.M. > G.M. for distinct real numbers

$$\therefore \frac{x+y}{2} > \sqrt{xy}, \frac{y+z}{2} > \sqrt{yz}, \frac{z+x}{2} > \sqrt{zx}$$

$$\Rightarrow \left(\frac{x+y}{2}\right)\left(\frac{y+z}{2}\right)\left(\frac{z+x}{2}\right) > xyz$$

$$\Rightarrow abc > (a+b-c)(b+c-a)(c+a-b)$$

$$\Rightarrow (b+c-a)(c+a-b)(a+b-c) - abc < 0$$

Case II : If $a+b-c = -ve$ then

$$(b+c-a)(c+a-b)(a+b-c) - abc$$

$$= (+ve)(+ve)(-ve) - (+ve)$$

$$= (-ve) - (+ve) = (-ve)$$

$$\Rightarrow (b+c-a)(c+a-b)(a+b-c) < abc$$

Hence in either case given expression is $-ve$.

4. (b) Given that a, b, c, d, p are real and distinct numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0$$

$$\Rightarrow (a^2 p^2 + b^2 p^2 + c^2 p^2) - (2abp + 2bcp + 2cdp) + (b^2 + c^2 + d^2) \leq 0$$

$$\Rightarrow (a^2 p^2 - 2abp + b^2) + (b^2 p^2 - 2bcp + c^2) + (c^2 p^2 - 2cdp + d^2) \leq 0$$

$$\Rightarrow (ap-b)^2 + (bp-c)^2 + (cp-d)^2 \leq 0$$

Being sum of perfect squares, LHS can never be $-ve$, therefore the only possibility is

$$(ap-b)^2 + (bp-c)^2 + (cp-d)^2 = 0$$

Which is possible only when each term is zero individually i.e.

$$ap-b=0; bp-c=0; cp-d=0$$

$$\Rightarrow \frac{b}{a} = p; \frac{c}{b} = p; \frac{d}{c} = p \Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c} = p$$

$$\Rightarrow a, b, c, d \text{ are in G.P.}$$

5. (a, b, c) The given equation is, $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$
For $x > 0$, taking log on both sides to the base x , we get

$$\frac{3}{4}(\log_2 x)^2 + (\log_2 x) - \frac{5}{4} = \log_x \sqrt{2} = \frac{1}{2} \log_x 2$$

$$\text{Let } \log_2 x = y, \text{ then we get, } \frac{3}{4}y^2 + y - \frac{5}{4} = \frac{1}{2y}$$

$$\Rightarrow 3y^3 + 4y^2 - 5y - 2 = 0$$

$$\Rightarrow (y-1)(y+2)(3y+1) = 0 \Rightarrow y = 1, -2, -1/3$$

$$\Rightarrow \log_2 x = 1, -2, -1/3 \Rightarrow x = 2, 2^{-2}, 2^{-1/3}$$

$$\Rightarrow x = 2, \frac{1}{4}, \frac{1}{2^{1/3}} \quad (\text{All accepted as } > 0)$$

$\therefore$ There are three real solution in which one is irrational.

6. (d) Let $x_1, x_2, \dots, x_n$ be the n +ve numbers

According to the question,

$$x_1 x_2 x_3 \dots x_n = 1 \quad \dots (1)$$

We know for +ve no.'s A.M. $\geq$ G.M.

$$\Rightarrow \frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

$$\Rightarrow \frac{x_1 + x_2 + \dots + x_n}{n} \geq 1 \quad [\text{Using eq. (1)}]$$

$$\Rightarrow x_1 + x_2 + \dots + x_n \geq n$$

7. (a) We have $240 = 2^4 \cdot 3 \cdot 5$.
Divisors of 240 are

$$1, 2, \quad 4, 8, \quad 16$$

$$3, 6, \quad 12, 24, \quad 48$$

$$5, 10, \quad 20, 40, \quad 80,$$

$$15, 30, \quad 60, 120, \quad 240$$

Out of these divisors just 4 divisors viz., 2, 6, 10, 30 are of the form $4n+2$.

8. (a, b, c)

$$3^x = 4^{x-1} \Rightarrow x \log 3 = 2(x-1) \log 2$$

$$\Rightarrow x = \frac{2 \log 2}{2 \log 2 - \log 3} \Rightarrow x = \frac{2 \log_3 2}{2 \log_3 2 - 1} = \frac{2}{2 - \log_2 3}$$

$$\text{Also } x = \frac{1}{1 - \frac{1}{2} \log_2 3} = \frac{1}{1 - \log_4 3}$$

9. (a, d) $\alpha x^2 - x + \alpha = 0$ has distinct real roots.

$$\therefore D > 0 \Rightarrow 1 - 4\alpha^2 > 0$$

$$\Rightarrow \alpha \in \left(-\frac{1}{2}, \frac{1}{2}\right) \quad \dots (i)$$

$$\text{Also } |x_1 - x_2| < 1$$

$$\Rightarrow (x_1 - x_2)^2 < 1 \Rightarrow (x_1 + x_2)^2 - 4x_1 x_2 < 1$$

$$\Rightarrow \frac{1}{\alpha^2} - 4 < 1 \Rightarrow \frac{1}{\alpha^2} < 5 \text{ or } \alpha^2 > \frac{1}{5}$$

$$\Rightarrow \alpha \in \left(-\infty, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right) \quad \dots (ii)$$

Combining (i) and (ii)

$$S = \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$$

$$\therefore \text{Subsets of } S \text{ can be } \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \text{ and } \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right).$$

E. Subjective Problems

1. $4^x - 3^{x-1/2} = 3^{x+1/2} - \frac{(2^2)^x}{2}$

$$\Rightarrow 4^x - \frac{3^x}{\sqrt{3}} = 3^x \sqrt{3} - \frac{4^x}{2}$$

$$\Rightarrow \frac{3}{2} \cdot 4^x = 3^x \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right) \Rightarrow \frac{3}{2} \cdot 4^x = 3^x \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{4^{x-1}}{4^{1/2}} = \frac{3^{x-1}}{\sqrt{3}} \Rightarrow 4^{x-3/2} = 3^{x-3/2}$$

Quadratic Equation and Inequalities (Inequalities)

$$\Rightarrow \left(\frac{4}{3}\right)^{x-3/2} = 1 \Rightarrow x - \frac{3}{2} = 0 \Rightarrow x = 3/2$$

$$\begin{aligned} 2. \text{ RHS} &= (m-1, n+1) + x^{m-n-1}(m-1, n) \\ &= \frac{(1-x^{m-1})(1-x^{m-2})\dots(1-x^{m-n-1})}{(1-x)(1-x^2)\dots(1-x^{n+1})} \\ &\quad + x^{m-n-1} \left[\frac{(1-x^{m-1})(1-x^{m-2})\dots(1-x^{m-n})}{(1-x)(1-x^2)\dots(1-x^n)} \right] \\ &= \frac{(1-x^{m-1})(1-x^{m-2})\dots(1-x^{m-n})}{(1-x)(1-x^2)\dots(1-x^n)} \\ &\quad \left[\frac{1-x^{m-n-1}}{1-x^{n+1}} + x^{m-n-1} \right] \\ &= \frac{1-x^{m-n-1} + x^{m-n-1} - x^m}{1-x^{n+1}} \\ &= \frac{(1-x^m)(1-x^{m-1})\dots(1-x^{m-n})}{(1-x)(1-x^2)\dots(1-x^n)(1-x^{n+1})} \\ &= (m, n+1) = \text{L.H.S.} \quad \text{Hence Proved} \end{aligned}$$

$$\begin{aligned} 3. \quad \sqrt{x+1} &= 1 + \sqrt{x-1} \\ \text{Squaring both sides, we get} \\ x+1 &= 1+x-1+2\sqrt{x-1} \Rightarrow 1=2\sqrt{x-1} \\ \Rightarrow 1 &= 4(x-1) \\ \Rightarrow x &= 5/4 \end{aligned}$$

$$4. \text{ Given } a > 0, \text{ so we have to consider two cases :}$$

$a \neq 1$ and $a = 1$. Also it is clear that $x > 0$ and $x \neq 1, ax \neq 1, a^2x \neq 1$.

Case I : If $a > 0, \neq 1$
then given equation can be simplified as

$$\frac{2}{\log_a x} + \frac{1}{1+\log_a x} + \frac{3}{2+\log_a x} = 0$$

Putting $\log_a x = y$, we get

$$2(1+y)(2+y) + y(2+y) + 3y(1+y) = 0$$

$$\Rightarrow 6y^2 + 11y + 4 = 0 \Rightarrow y = -4/3 \text{ and } -1/2$$

$$\Rightarrow \log_a x = -4/3 \text{ and } \log_a x = -1/2$$

$$\Rightarrow x = a^{-4/3} \text{ and } x = a^{-1/2}$$

Case II : If $a = 1$ then equation becomes

$$2 \log_x 1 + \log_x 1 + 3 \log_x 1 = 6 \log_x 1 = 0$$

which is true $\forall x > 0, \neq 1$

Hence solution is if $a = 1, x > 0, \neq 1$

if $a > 0, \neq 1 ; x = a^{-1/2}, a^{-4/3}$

$$5. \text{ Let } x = \frac{\sqrt{26-15\sqrt{3}}}{5\sqrt{2}-\sqrt{38+5\sqrt{3}}}$$

$$\Rightarrow x^2 = \frac{26-15\sqrt{3}}{50+38+5\sqrt{3}-10\sqrt{76+10\sqrt{3}}}$$

$$\Rightarrow x^2 = \frac{26-15\sqrt{3}}{88+5\sqrt{3}-10\sqrt{75+1+10\sqrt{3}}}$$

$$\Rightarrow x^2 = \frac{26-15\sqrt{3}}{88+5\sqrt{3}-10\sqrt{(5\sqrt{3})^2+(1)^2+2 \times 5\sqrt{3} \times 1}}$$

$$\begin{aligned} \Rightarrow x^2 &= \frac{26-15\sqrt{3}}{88+5\sqrt{3}-10\sqrt{(5\sqrt{3}+1)^2}} \\ &= \frac{26-15\sqrt{3}}{3(26-15\sqrt{3})} = \frac{1}{3}, \text{ which is a rational number.} \end{aligned}$$

$$6. \quad x^2 + y^2 - 2x \geq 0 \Rightarrow x^2 - 2x + 1 + y^2 \geq 1$$

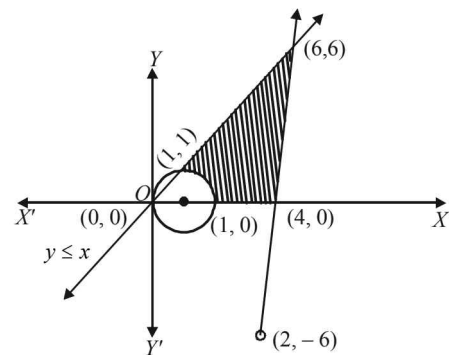
$$\Rightarrow (x-1)^2 + y^2 \geq 1 \text{ which represents the boundary and exterior region of the circle with centre at } (1,0) \text{ and radius as } 1.$$

For $3x - y \leq 12$, the corresponding equation is $3x - y = 12$; any two points on it can be taken as $(4, 0), (2, -6)$. Also putting $(0, 0)$ in given inequation, we get $0 \leq 12$ which is true.

$\therefore$ given inequation represents that half plane region of line $3x - y = 12$ which contains origin.

For $y \leq x$, the corresponding equation $y = x$ has any two points on it as $(0, 0)$ and $(1, 1)$. Also putting $(2, 1)$ in the given inequation, we get $1 \leq 2$ which is true, so $y \leq x$ represents that half plane which contains the points $(2, 1)$. $y \geq 0$ represents upper half cartesian plane.

Combining all we find the solution set as the shaded region in the graph.



$$7. \text{ There are two parts of this question}$$

$$(5x-1) < (x+1)^2 \text{ and } (x+1)^2 < (7x-3)$$

Taking first part

$$(5x-1) < (x+1)^2 \Rightarrow 5x-1 < x^2+2x+1$$

$$\Rightarrow x^2-3x+2 > 0 \Rightarrow (x-1)(x-2) > 0$$

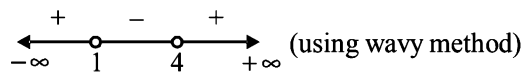
$$\begin{array}{c} + \quad - \quad + \\ \leftarrow \quad \circ \quad 1 \quad \circ \quad 2 \quad \rightarrow \end{array} \text{ (using wavy method)}$$

$$\Rightarrow x < 1 \text{ or } x > 2 \quad \dots(1)$$

Taking second part

$$(x+1)^2 < (7x-3) \Rightarrow x^2 - 5x + 4 < 0$$

$$\Rightarrow (x-1)(x-4) < 0$$



$$\Rightarrow 1 < x < 4 \quad \dots(2)$$

Combining (1) and (2) [taking common solution], we get $2 < x < 4$ but x is an integer therefore $x = 3$.

8. $\therefore \alpha, \beta$ are the roots of $x^2 + px + q = 0$

$$\therefore \alpha + \beta = -p, \quad \alpha\beta = q$$

$$\therefore \gamma, \delta \text{ are the roots of } x^2 + rx + s = 0$$

$$\therefore \gamma + \delta = -r, \gamma\delta = s$$

$$\text{Now, } (\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$$

$$= [\alpha^2 - (\gamma + \delta)\alpha + \gamma\delta][\beta^2 - (\gamma + \delta)\beta + \gamma\delta]$$

$$= [\alpha^2 + r\alpha + s][\beta^2 + r\beta + s] \quad [\because \alpha, \beta \text{ are roots of } x^2 + px + q = 0]$$

$$\therefore \alpha^2 + p\alpha + q = 0 \text{ and } \beta^2 + p\beta + q = 0$$

$$= [(r-p)\alpha + (s-q)][(r-p)\beta + (s-q)]$$

$$= (r-p)^2\alpha\beta + (r-p)(s-q)(\alpha + \beta) + (s-q)^2$$

$$= q(r-p)^2 - p(r-p)(s-q) + (s-q)^2$$

Now if the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ have a common root say α , then $\alpha^2 + p\alpha + q = 0$ and $\alpha^2 + r\alpha + s = 0$

$$\Rightarrow \frac{\alpha^2}{ps - qr} = \frac{\alpha}{q - s} = \frac{1}{r - p}$$

$$\Rightarrow \alpha^2 = \frac{ps - qr}{r - p} \text{ and } \alpha = \frac{q - s}{r - p}$$

$$\Rightarrow (q - s)^2 = (r - p)(ps - qr) \text{ which is the required condition.}$$

9. Given that $n^4 < 10^n$ for a fixed +ve integer $n \geq 2$.

To prove that $(n+1)^4 < 10^{n+1}$

Proof: Since $n^4 < 10^n \Rightarrow 10n^4 < 10^{n+1} \quad \dots(1)$

So it is sufficient to prove that $(n+1)^4 < 10n^4$

$$\text{Now } \left(\frac{n+1}{n}\right)^4 = \left(1 + \frac{1}{n}\right)^4 \leq \left(1 + \frac{1}{2}\right)^4 \quad [\because n \geq 2]$$

$$= \frac{81}{16} < 10$$

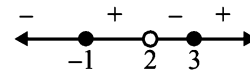
$$\Rightarrow (n+1)^4 < 10n^4 \quad \dots(2)$$

From (1) and (2), $(n+1)^4 < 10^{n+1}$

10. $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$

y will take all real values if $\frac{(x+1)(x-3)}{(x-2)} \geq 0$

By wavy method



$$x \in [-1, 2) \cup [3, \infty)$$

[2 is not included as it makes denominator zero, and hence y an undefined number.]

11. The given equations are $3x + my - m = 0$ and $2x - 5y - 20 = 0$. Solving these equations by cross product method, we get

$$\frac{x}{-20m - 5m} = \frac{y}{-2m + 60} = \frac{1}{-15 - 2m} \quad \text{NOTE THIS STEP}$$

$$\Rightarrow x = \frac{25m}{2m+15}, y = \frac{2m-60}{2m+15}$$

$$\text{For } x > 0 \Rightarrow \frac{25m}{2m+15} > 0$$

$$\Rightarrow m < -\frac{15}{2} \text{ or } m > 0 \quad \dots(1)$$

$$\text{For } y > 0 \Rightarrow \frac{2(m-30)}{2m+15} > 0$$

$$\Rightarrow m < -\frac{15}{2} \text{ or } m > 30 \quad \dots(2)$$

Combining (1) and (2), we get the common values of m as follows:

$$m < -\frac{15}{2} \text{ or } m > 30 \quad \therefore m \in \left(-\infty, -\frac{15}{2}\right) \cup (30, \infty)$$

12. The given system is

$$x + 2y + z = 1 \quad \dots(1)$$

$$2x - 3y - \omega = 2 \quad \dots(2)$$

where $x, y, z, \omega \geq 0$

Multiplying eqn. (1) by 2 and subtracting from (2), we get

$$7y + 2z + \omega = 0 \Rightarrow \omega = -(7y + 2z)$$

Now if $y, z > 0, \omega < 0$ (not possible)

If $y = 0, z = 0$ then $x = 1$ and $\omega = 0$.

$\therefore$ The only solution is $x = 1, y = 0, z = 0, \omega = 0$.

13. $e^{\sin x} - e^{-\sin x} - 4 = 0$

Let $e^{\sin x} = y$ then $e^{-\sin x} = 1/y$

$$\therefore \text{Equation becomes, } y - \frac{1}{y} - 4 = 0$$

$$\Rightarrow y^2 - 4y - 1 = 0 \Rightarrow y = 2 + \sqrt{5}, 2 - \sqrt{5}$$

But y is real +ve number,

$$\therefore y \neq 2 - \sqrt{5} \Rightarrow y = 2 + \sqrt{5}$$

$$\Rightarrow e^{\sin x} = 2 + \sqrt{5} \Rightarrow \sin x = \log_e(2 + \sqrt{5})$$

$$\text{But } 2 + \sqrt{5} > e \Rightarrow \log_e(2 + \sqrt{5}) > \log_e e$$

Quadratic Equation and Inequalities (Inequalities)

$$\Rightarrow \log_e(2 + \sqrt{5}) > 1 \quad \text{Hence, } \sin x > 1$$

Which is not possible.

$\therefore$ Given equation has no real solution.

14. For any square there can be at most 4, neighbouring squares.

		q				
	P	d	r			
		s				

Let for a square having largest number d, p, q, r, s be written then

According to the question,

$$p + q + r + s = 4d$$

$$\Rightarrow (d-p) + (d-q) + (d-r) + (d-s) = 0$$

Sum of four +ve numbers can be zero only if these are zero individually

$$\therefore d-p=0=d-q=d-r=d-s$$

$$\Rightarrow p=q=r=s=d$$

$\Rightarrow$ all the numbers written are same.

Hence Proved.

15. Let α, β be the roots of eq. $ax^2 + bx + c = 0$

According to the question, $\beta = \alpha^n$

$$\text{Also } \alpha + \beta = -b/a ; \alpha\beta = c/a$$

$$\alpha\beta = \frac{c}{a} \Rightarrow \alpha \cdot \alpha^n = \frac{c}{a} \Rightarrow \alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}}$$

$$\text{then } \alpha + \beta = -b/a \Rightarrow \alpha + \alpha^n = \frac{-b}{a}$$

$$\text{or } \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + \left(\frac{c}{a}\right)^{\frac{n}{n+1}} = \frac{-b}{a}$$

$$\Rightarrow a \cdot \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + a \cdot \left(\frac{c}{a}\right)^{\frac{n}{n+1}} + b = 0$$

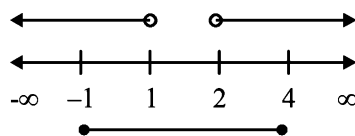
$$\Rightarrow a^{\frac{n}{n+1}} c^{\frac{1}{n+1}} + a^{\frac{1}{n+1}} c^{\frac{n}{n+1}} + b = 0$$

$$\Rightarrow (a^n c)^{\frac{1}{n+1}} + (ac^n)^{\frac{1}{n+1}} + b = 0$$

Hence Proved.

16. $x^2 - 3x + 2 > 0$, $x^2 - 3x - 4 \leq 0$

$$\Rightarrow (x-1)(x-2) > 0 \text{ and } (x-4)(x+1) < 0$$



$$\Rightarrow x \in (-\infty, 1) \cup (2, \infty) \text{ and } x \in [-1, 4]$$

$\therefore$ Common solution is $[-1, 1) \cup (2, 4]$

17. The given equation is

$$(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10 \quad \dots(1)$$

$$\text{Let } (5 + 2\sqrt{6})^{x^2-3} = y \quad \dots(2)$$

$$\begin{aligned} \text{then } (5 - 2\sqrt{6})^{x^2-3} &= \left(\frac{(5 - 2\sqrt{6})(5 + 2\sqrt{6})}{5 + 2\sqrt{6}} \right)^{x^2-3} \\ &= \left(\frac{25 - 24}{5 + 2\sqrt{6}} \right)^{x^2-3} = \left(\frac{1}{5 + 2\sqrt{6}} \right)^{x^2-3} = \frac{1}{y} \text{ (Using (2))} \end{aligned}$$

$$\therefore \text{ The given equation (1) becomes } y + \frac{1}{y} = 10$$

$$\Rightarrow y^2 - 10y + 1 = 0 \Rightarrow y = \frac{10 \pm \sqrt{100 - 4}}{2} = \frac{10 \pm 4\sqrt{6}}{2}$$

$$\Rightarrow y = 5 + 2\sqrt{6} \text{ or } 5 - 2\sqrt{6}$$

Consider, $y = 5 + 2\sqrt{6}$

$$\Rightarrow (5 + 2\sqrt{6})^{x^2-3} = (5 + 2\sqrt{6})$$

$$\Rightarrow x^2 - 3 = 1 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$$

Again consider

$$y = 5 - 2\sqrt{6} = \frac{1}{5 + 2\sqrt{6}} = (5 + 2\sqrt{6})^{-1}$$

$$\Rightarrow (5 + 2\sqrt{6})^{x^2-3} = (5 + 2\sqrt{6})^{-1} \Rightarrow x^2 - 3 = -1$$

$$\Rightarrow x^2 = 2 \Rightarrow x = \pm \sqrt{2}$$

Hence the solutions are $2, -2, \sqrt{2}, -\sqrt{2}$.

18. The given equation is,

$$x^2 - 2a|x - a| - 3a^2 = 0$$

Here two cases are possible.

Case I: $x - a > 0$ then $|x - a| = x - a$

$\therefore$ Eq. becomes

$$x^2 - 2a(x - a) - 3a^2 = 0$$

$$\text{or } x^2 - 2ax - a^2 = 0 \Rightarrow x = \frac{2a \pm \sqrt{4a^2 + 4a^2}}{2}$$

$$\Rightarrow x = a \pm a\sqrt{2}$$

Case II: $x - a < 0$ then $|x - a| = -(x - a)$

$\therefore$ Eq. becomes

$$x^2 + 2a(x - a) - 3a^2 = 0$$

$$\text{or } x^2 + 2ax - 5a^2 = 0 \Rightarrow x = \frac{-2a \pm \sqrt{4a^2 + 20a^2}}{2}$$

$$\Rightarrow x = \frac{-2a \pm 2a\sqrt{6}}{2} \Rightarrow x = -a \pm a\sqrt{6}$$

Thus the solution set is $\{a \pm a\sqrt{2}, -a \pm a\sqrt{6}\}$

19. We are given $\frac{2x}{2x^2 + 5x + 2} > \frac{1}{x+1}$

$$\Rightarrow \frac{2x}{2x^2 + 5x + 2} - \frac{1}{x+1} > 0$$

$$\Rightarrow \frac{2x^2 + 2x - 2x^2 - 5x - 2}{(2x^2 + 5x + 2)(x+1)} > 0$$

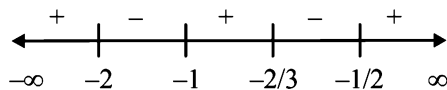
$$\Rightarrow \frac{-3x-2}{(2x+1)(x+1)(x+2)} > 0 \Rightarrow \frac{(3x+2)}{(x+1)(x+2)(2x+1)} < 0$$

$$\Rightarrow \frac{(3x+2)(x+1)(x+2)(2x+1)}{(x+1)^2(x+2)^2(2x+1)^2} < 0$$

$$\Rightarrow (3x+2)(x+1)(x+2)(2x+1) < 0 \quad \dots(1)$$

NOTE THIS STEP: Critical pts are $x = -2/3, -1, -2, -1/2$

On number line



Clearly Inequality (1) holds for,

$$x \in (-2, -1) \cup (-2/3, -1/2)$$

$$[as \ x \neq -2, -1, -2/3, -1/2]$$

20. The Given equation is,

$$|x^2 + 4x + 3| + 2x + 5 = 0$$

Now there can be two cases.

Case I: $x^2 + 4x + 3 \geq 0 \Rightarrow (x+1)(x+3) \geq 0$

$$\Rightarrow x \in (-\infty, -3] \cup [-1, \infty) \quad \dots(i)$$

Then given equation becomes,

$$\Rightarrow x^2 + 6x + 8 = 0$$

$$\Rightarrow (x+4)(x+2) = 0 \Rightarrow x = -4, -2$$

But $x = -2$ does not satisfy (i), hence rejected

$\therefore x = -4$ is the sol.

Case II: $x^2 + 4x + 3 < 0$

$$\Rightarrow (x+1)(x+3) < 0$$

$$\Rightarrow x \in (-3, -1) \quad \dots(ii)$$

Then given equation becomes,

$$-(x^2 + 4x + 3) + 2x + 5 = 0$$

$$\Rightarrow -x^2 - 2x + 2 = 0 \Rightarrow x^2 + 2x - 2 = 0$$

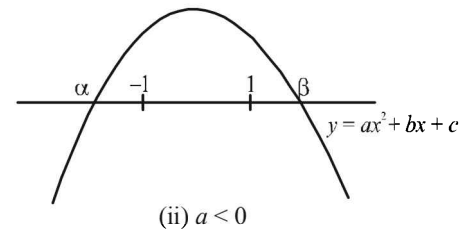
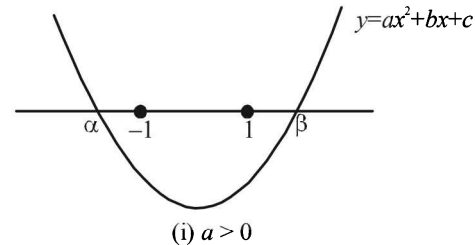
$$\Rightarrow x = \frac{-2 \pm \sqrt{4+8}}{2} \Rightarrow x = -1 + \sqrt{3}, -1 - \sqrt{3}$$

Out of which $x = -1 - \sqrt{3}$ is sol.

Combining the two cases we get the solutions of given equation as $x = -4, -1 - \sqrt{3}$.

21. Given that for $a, b, c \in R$, $ax^2 + bx + c = 0$ has two real roots α and β , where $\alpha < -1$ and $\beta > 1$. There may be two cases depending upon value of a , as shown below.

In each of cases (i) and (ii) $af(-1) < 0$ and $af(1) < 0$



$$\Rightarrow a(a-b+c) < 0 \text{ and } a(a+b+c) < 0$$

Dividing by $a^2 (> 0)$, we get

$$1 - \frac{b}{a} + \frac{c}{a} < 0 \quad \dots(1)$$

and $1 + \frac{b}{a} + \frac{c}{a} < 0 \quad \dots(2)$

Combining (1) and (2) we get

$$1 + \left| \frac{b}{a} \right| + \frac{c}{a} < 0 \text{ or } 1 + \frac{c}{a} + \left| \frac{b}{a} \right| < 0 \text{ Hence Proved.}$$

22. $a^2 = p^2 + s^2, b^2 = (1-p)^2 + q^2$
 $c^2 = (1-q)^2 + (1-r)^2, d^2 = r^2 + (1-s)^2$
 $\therefore a^2 + b^2 + c^2 + d^2 = \{p^2 + (1-p)^2\} + \{q^2 + (1-q)^2\}$
 $+ \{r^2 + (1-r)^2\} + \{s^2 + (1-s)^2\}$

where p, q, r, s all vary in the interval $[0, 1]$.

Now consider the function

$$y^2 = x^2 + (1-x)^2, 0 \leq x \leq 1,$$

$$2y \frac{dy}{dx} = 2x - 2(1-x) = 0$$

$$\Rightarrow x = \frac{1}{2} \text{ which } \frac{d^2y}{dx^2} = 4 \text{ i.e. +ive}$$

Hence y is minimum at $x = \frac{1}{2}$ and its minimum

$$\text{value is } \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

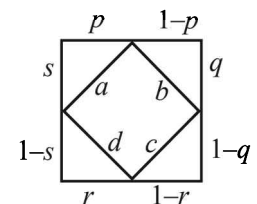
Clearly value is maximum at the end pts which is 1.

$$\therefore \text{Minimum value of } a^2 + b^2 + c^2 + d^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2$$

and maximum value is $1 + 1 + 1 + 1 = 4$. Hence proved.

23. We know that,

$$(\alpha - \beta)^2 = [(\alpha + \delta) - (\beta + \delta)]^2$$



Quadratic Equation and Inequalities (Inequalities)

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\alpha + \delta + \beta + \delta)^2 - 4(\alpha + \delta)(\beta + \delta)$$

$$\Rightarrow \frac{b^2}{a^2} - \frac{4c}{a} = \frac{B^2}{A^2} - \frac{4C}{A} \Rightarrow \frac{4ac - b^2}{a^2} = \frac{4AC - B^2}{A^2}$$

$$[\text{Here } \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a},$$

$$(\alpha + \delta)(\beta + \delta)$$

$$= -\frac{B}{A} \text{ and } (\alpha + \delta)(\beta + \delta) = \frac{C}{A}]$$

Hence proved.

24. Divide the equation by a^3 , we get

$$x^2 + \frac{b}{a} \cdot \frac{c}{a} x + \left(\frac{c}{a}\right)^3 = 0$$

$$\Rightarrow x^2 - (\alpha + \beta) \cdot (\alpha\beta)x + (\alpha\beta)^3 = 0$$

$$\Rightarrow x^2 - \alpha^2\beta x - \alpha\beta^2 x + (\alpha\beta)^3 = 0$$

$$\Rightarrow x(x - \alpha^2\beta) - \alpha\beta^2(x - \alpha^2\beta) = 0$$

$$\Rightarrow (x - \alpha^2\beta)(x - \alpha\beta^2) = 0$$

$$\Rightarrow x = \alpha^2\beta, \alpha\beta^2 \text{ which is the required answer.}$$

25. The given equation is,

$$x^2 + (a - b)x + (1 - a - b) = 0, a, b \in R$$

For this eqⁿ to have unequal real roots $\forall b$

$$D > 0$$

$$\Rightarrow (a - b)^2 - 4(1 - a - b) > 0$$

$$\Rightarrow a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$$

$$\Rightarrow b^2 + b(4 - 2a) + a^2 + 4a - 4 > 0$$

Which is a quadratic expression in b , and it will be true

$\forall b \in R$ if discriminant of above eqⁿ less than zero.

$$\text{i.e., } (4 - 2a)^2 - 4(a^2 + 4a - 4) < 0$$

$$\Rightarrow (2 - a)^2 - (a^2 + 4a - 4) < 0$$

$$\Rightarrow 4 - 4a + a^2 - a^2 - 4a + 4 < 0$$

$$\Rightarrow -8a + 8 < 0$$

$$\Rightarrow a > 1$$

26. Given that a, b, c are positive real numbers. To prove that

$$(a + 1)^7(b + 1)^7(c + 1)^7 > 7^7 a^4 b^4 c^4$$

$$\text{Consider L.H.S.} = (1 + a)^7 \cdot (1 + b)^7 \cdot (1 + c)^7$$

$$= [(1 + a)(1 + b)(1 + c)]^7$$

$$[1 + a + b + c + ab + bc + ca + abc]^7$$

$$> [a + b + c + ab + bc + ca + abc]^7 \quad \dots(1)$$

Now we know that AM $\geq$ GM using it for +ve no's a, b, c, ab, bc, ca and abc , we get

$$\frac{a + b + c + ab + bc + ca + abc}{7} \geq (a^4 b^4 c^4)^{1/7}$$

$$\Rightarrow (a + b + c + ab + bc + ca + abc)^7 \geq 7^7 (a^4 b^4 c^4)$$

From (1) and (2), we get

$$[(1 + a)(1 + b)(1 + c)]^7 > 7^7 a^4 b^4 c^4$$

Hence Proved.

27. Roots of $x^2 - 10cx - 11d = 0$ are a and b

$$\Rightarrow a + b = 10c \text{ and } ab = -11d$$

Similarly c and d are the roots of $x^2 - 10ax - 11b = 0$

$$\Rightarrow c + d = 10a \text{ and } cd = -11b$$

$$\Rightarrow a + b + c + d = 10(a + c) \text{ and } abcd = 121bd$$

$$\Rightarrow b + d = 9(a + c) \text{ and } ac = 121$$

Also we have $a^2 - 10ac - 11d = 0$ and $c^2 - 10ac - 11b = 0$

$$\Rightarrow a^2 + c^2 - 20ac - 11(b + d) = 0$$

$$\Rightarrow (a + c)^2 - 22 \times 121 - 99(a + c) = 0$$

$$\Rightarrow a + c = 121 \text{ or } -22$$

For $a + c = -22$, we get $a = c$

$\therefore$ rejecting this value we have $a + c = 121$

$$\therefore a + b + c + d = 10(a + c) = 1210$$

H. Assertion & Reason Type Questions

1. (b) As $a, b, c, p, q, \in R$ and the two given equations have exactly one common root

$\Rightarrow$ Either both equations have real roots

or both equations have imaginary roots

$\Rightarrow$ Either $\Delta_1 \geq 0$ and $\Delta_2 \geq 0$ or $\Delta_1 \leq 0$ and $\Delta_2 \leq 0$

$$\Rightarrow p^2 - q \geq 0 \text{ and } b^2 - ac \geq 0$$

$$\text{or } p^2 - q \leq 0 \text{ and } b^2 - ac \leq 0$$

$$\Rightarrow (p^2 - q)(b^2 - ac) \geq 0$$

$\therefore$ Statement 1 is true.

$$\text{Also we have } \alpha\beta = q \text{ and } \frac{\alpha}{\beta} = \frac{c}{a}$$

$$\therefore \frac{\alpha\beta}{\alpha/\beta} = \frac{q}{c} \times a \Rightarrow \beta^2 = \frac{qa}{c}$$

$$\text{As } \beta \neq 1 \text{ or } -1 \Rightarrow \beta^2 \neq 1 \Rightarrow \frac{qa}{c} \neq 1 \text{ or } c \neq qa$$

Again, as exactly one root α is common, and $\beta \neq 1$

$$\therefore \alpha + \beta \neq \alpha + \frac{1}{\beta} \Rightarrow \frac{-2b}{a} \neq -2p \Rightarrow b \neq ap$$

$\therefore$ Statement 2 is correct.

But Statement 2 is not a correct explanation of Statement 1.

I. Integer Value Correct Type

1. (7) The given system of equations is

$$3x - y - z = 0$$

$$-3x + z = 0$$

$$-3x + 2y + z = 0$$

Let $x = p$ where p is an integer, then $y = 0$ and $z = 3p$

$$\text{But } x^2 + y^2 + z^2 \leq 100 \Rightarrow p^2 + 9p^2 \leq 100$$

$$\Rightarrow p^2 \leq 10 \Rightarrow p = 0, \pm 1, \pm 2, \pm 3$$

i.e. p can take 7 different values.

$\therefore$ Number of points (x, y, z) are 7.

2. (2) The given equation is

$$x^2 - 8kx + 16(k^2 - k + 1) = 0$$

$\therefore$ Both the roots are real and distinct

$$\therefore D > 0 \Rightarrow (8k)^2 - 4 \times 16(k^2 - k + 1) > 0$$

$$\Rightarrow k > 1$$

...(i)

$\therefore$ Both the roots are greater than or equal to 4

$\therefore \alpha + \beta > 8$ and $f(4) \geq 0$

$\Rightarrow k > 1$... (ii)

and $16 - 32k + 16(k^2 - k + 1) \geq 0$

$\Rightarrow k^2 - 3k + 2 \geq 0 \Rightarrow (k-1)(k-2) \geq 0$

$\Rightarrow k \in (-\infty, 1] \cup [2, \infty)$... (iii)

Combining (i), (ii) and (iii), we get $k \geq 2$ or the smallest value of $k = 2$.

3. (8) $\therefore a > 0, \therefore a^{-5}, a^{-4}, 3a^{-3}, 1, a^8, a^{10} > 0$

Using $AM \geq GM$ for positive real numbers we get

$$\frac{\frac{1}{a^5} + \frac{1}{a^4} + \frac{1}{a^3} + \frac{1}{a^3} + \frac{1}{a^3} + 1 + a^8 + a^{10}}{8} \geq$$

$$\left(\frac{1}{a^5} \cdot \frac{1}{a^4} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot 1 \cdot a^8 \cdot a^{10} \right)^{\frac{1}{8}}$$

$$\Rightarrow \frac{1}{a^5} + \frac{1}{a^4} + \frac{3}{a^3} + 1 + a^8 + a^{10} \geq 8(1)^{\frac{1}{8}}$$

4. (2) We have $x^4 - 4x^3 + 12x^2 + x - 1 = 0$
 $\Rightarrow x^4 - 4x^3 + 6x^2 - 4x + 1 + 6x^2 + 5x - 2 = 0$
 $\Rightarrow (x-1)^4 + 6x^2 + 5x - 2 = 0$

$$\Rightarrow (x-1)^4 = -6x^2 - 5x + 2$$

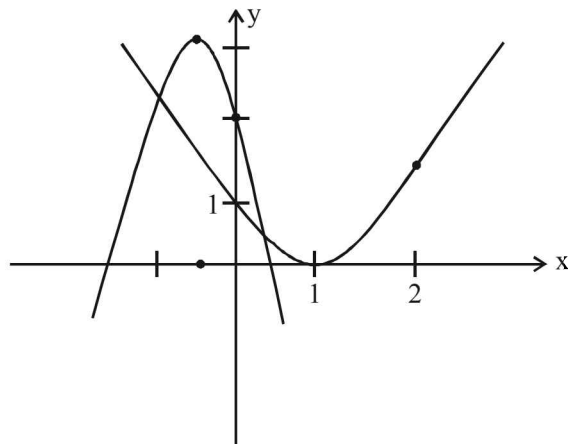
To solve the above polynomial, it is equivalent to find the intersection points of the curves $y = (x-1)^4$ and

$$y = -6x^2 - 5x + 2 \text{ or } y = (x-1)^4 \text{ and } \left(x + \frac{5}{12}\right)^2 = -\frac{1}{6}\left(y - \frac{73}{24}\right)$$

The graph of above two curves as follows.

Clearly they have two points of intersection.

Hence the given polynomial has two real roots.



Section-B

JEE Main/ AIEEE

1. (a) We have $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$;
 $\Rightarrow \alpha$ & β are roots of equation, $x^2 = 5x - 3$
or $x^2 - 5x + 3 = 0$ $\therefore \alpha + \beta = 5$ and $\alpha\beta = 3$

Thus, the equation having $\frac{\alpha}{\beta}$ & $\frac{\beta}{\alpha}$ as its roots is

$$x^2 - x\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + \frac{\alpha\beta}{\alpha\beta} = 0$$

$$\Rightarrow x^2 - x\left(\frac{\alpha^2 + \beta^2}{\alpha\beta}\right) + 1 = 0 \text{ or } 3x^2 - 19x + 3 = 0$$

2. (a) Let α, β and γ, δ be the roots of the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ respectively.

$\therefore \alpha + \beta = -a, \alpha\beta = b$ and $\gamma + \delta = -b, \gamma\delta = a$.

Given $|\alpha - \beta| = |\gamma - \delta| \Rightarrow (\alpha - \beta)^2 = (\gamma - \delta)^2$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$$

$$\Rightarrow a^2 - 4b = b^2 - 4a \Rightarrow (a^2 - b^2) + 4(a - b) = 0$$

$$\Rightarrow a + b + 4 = 0 \quad (\because a \neq b)$$

3. (a) Product of real roots $= \frac{9}{t^2} > 0, \forall t \in \mathbb{R}$

$\therefore$ Product of real roots is always positive.

4. (a) $p + q = -p$ and $pq = q \Rightarrow q(p-1) = 0$

$$\Rightarrow q = 0 \text{ or } p = 1.$$

If $q = 0$, then $p = 0$. i.e. $p = q$

$$\therefore p = 1 \text{ and } q = -2.$$

5. (a) $\therefore (a-b)^2 + (b-c)^2 + (c-a)^2 > 0$
 $\Rightarrow 2(a^2 + b^2 + c^2 - ab - bc - ca) > 0$
 $\Rightarrow 2 > 2(ab + bc + ca) \Rightarrow ab + bc + ca < 1$

6. (d) $ax^2 + bx + c = 0, \alpha + \beta = \frac{-b}{a}, \alpha\beta = \frac{c}{a}$

$$\text{As for given condition, } \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\alpha + \beta = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} - \frac{b}{a} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c^2}{a^2}}$$

On simplification $2a^2c = ab^2 + bc^2$

$$\Rightarrow \frac{2a}{b} = \frac{c}{a} + \frac{b}{c} \Rightarrow \frac{c}{a}, \frac{a}{b}, \frac{b}{c} \text{ are in A.P.}$$

$\therefore \frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in H.P.

7. (b) Let the roots of given equation be α and 2α then

$$\alpha + 2\alpha = 3\alpha = \frac{1-3a}{a^2-5a+3}$$

$$\& \alpha \cdot 2\alpha = 2\alpha^2 = \frac{2}{a^2-5a+3} \Rightarrow \alpha = \frac{1-3a}{3(a^2-5a+3)}$$

Quadratic Equation and Inequalities (Inequalities)

$$\therefore 2 \left[\frac{1}{9} \frac{(1-3a)^2}{(a^2-5a+3)^2} \right] = \frac{2}{a^2-5a+3}$$

$$\frac{(1-3a)^2}{(a^2-5a+3)} = 9 \text{ or } 9a^2 - 6a + 1 = 9a^2 - 45a + 27$$

$$\text{or } 39a = 26 \text{ or } a = \frac{2}{3}$$

8. (c) $x^2 - 3|x| + 2 = 0 \Rightarrow |x|^2 - 3|x| + 2 = 0$
 $(|x| - 2)(|x| - 1) = 0$
 $|x| = 1, 2 \text{ or } x = \pm 1, \pm 2 \therefore \text{No. of solution} = 4$

9. (c) $y = x + \frac{1}{x} \text{ or } \frac{dy}{dx} = 1 - \frac{1}{x^2}$
 For max. or min., $1 - \frac{1}{x^2} = 0 \Rightarrow x = \pm 1$

$$\frac{d^2y}{dx^2} = \frac{2}{x^3} \Rightarrow \left(\frac{d^2y}{dx^2} \right)_{x=2} = 2 \text{ (+ve minima)} \therefore x = 1$$

10. (b) Let two numbers be a and b then $\frac{a+b}{2} = 9$ and

$$\sqrt{ab} = 4$$

$\therefore$ Equation with roots a and b is

$$x^2 - (a+b)x + ab = 0 \Rightarrow x^2 - 18x + 16 = 0$$

11. (c) Let the second root be α .

$$\text{Then } \alpha + (1-p) = -p \Rightarrow \alpha = -1$$

$$\text{Also } \alpha(1-p) = 1-p$$

$$\Rightarrow (\alpha - 1)(1-p) = 0 \Rightarrow p = 1 [\because \alpha = -1]$$

$$\therefore \text{Roots are } \alpha = -1 \text{ and } p - 1 = 0$$

12. (d) 4 is a root of $x^2 + px + 12 = 0$

$$\Rightarrow 16 + 4p + 12 = 0 \Rightarrow p = -7$$

Now, the equation $x^2 + px + q = 0$ has equal roots.

$$\therefore p^2 - 4q = 0 \Rightarrow q = \frac{p^2}{4} = \frac{49}{4}$$

13. (b) $\tan\left(\frac{P}{2}\right), \tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0$

$$\tan\left(\frac{P}{2}\right) + \tan\left(\frac{Q}{2}\right) = -\frac{b}{a}, \tan\left(\frac{P}{2}\right) \cdot \tan\left(\frac{Q}{2}\right) = \frac{c}{a}$$

$$\frac{\tan\left(\frac{P}{2}\right) + \tan\left(\frac{Q}{2}\right)}{1 - \tan\left(\frac{P}{2}\right)\tan\left(\frac{Q}{2}\right)} = \tan\left(\frac{P}{2} + \frac{Q}{2}\right) = 1$$

$$\Rightarrow \frac{-\frac{b}{a}}{1 - \frac{c}{a}} = 1 \Rightarrow -\frac{b}{a} = \frac{a-c}{a}$$

$$\Rightarrow -b = a - c \text{ or } c = a + b.$$

14. (c) both roots are less than 5

then (i) Discriminant ≥ 0

(ii) $p(5) > 0$

(iii) $\frac{\text{Sum of roots}}{2} < 5$

Hence (i) $4k^2 - 4(k^2 + k - 5) \geq 0$

$$4k^2 - 4k^2 - 4k + 20 \geq 0$$

$$4k \leq 20 \Rightarrow k \leq 5$$

(ii) $f(5) > 0; 25 - 10k + k^2 + k - 5 > 0$

$$\text{or } k^2 - 9k + 20 > 0$$

$$\text{or } k(k-4) - 5(k-4) > 0$$

$$\text{or } (k-5)(k-4) > 0$$

$$\Rightarrow k \in (-\infty, 4) \cup (-\infty, 5)$$

(ii) $\frac{\text{Sum of roots}}{2} = -\frac{b}{2a} = \frac{2k}{2} < 5$

The intersection of (i), (ii) & (iii) gives

$$k \in (-\infty, 4).$$

15. (b) $x^2 + px + q = 0$

$$\text{Sum of roots} = \tan 30^\circ + \tan 15^\circ = -p$$

$$\text{Product of roots} = \tan 30^\circ \cdot \tan 15^\circ = q$$

$$\tan 45^\circ = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \cdot \tan 15^\circ} = \frac{-p}{1-q} = 1$$

$$\Rightarrow -p = 1 - q \Rightarrow q - p = 1 \therefore 2 + q - p = 3$$

16. (c) Equation $x^2 - 2mx + m^2 - 1 = 0$

$$(x-m)^2 - 1 = 0 \text{ or } (x-m+1)(x-m-1) = 0$$

$$x = m-1, m+1$$

$$m-1 > -2 \text{ and } m+1 < 4$$

$$\Rightarrow m > -1 \text{ and } m < 3 \text{ or } -1 < m < 3$$

17. (b) $y = \frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$

$$3x^2(y-1) + 9x(y-1) + 7y - 17 = 0$$

$$D \geq 0 \because x \text{ is real}$$

$$81(y-1)^2 - 4 \times 3(y-1)(7y-17) \geq 0$$

$$\Rightarrow (y-1)(y-41) \leq 0 \Rightarrow 1 \leq y \leq 41$$

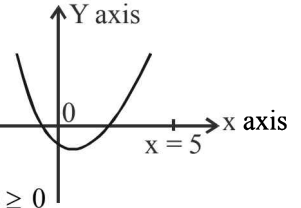
$\therefore$ Max value of y is 41

18. (c) Let α and β are roots of the equation

$$x^2 + ax + 1 = 0 \text{ So, } \alpha + \beta = -a \text{ and } \alpha\beta = 1$$

$$\text{given } |\alpha - \beta| < \sqrt{5} \Rightarrow \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} < \sqrt{5}$$

$$(\because (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta)$$



$$\Rightarrow \sqrt{a^2 - 4} < \sqrt{5} \Rightarrow a^2 - 4 < 5$$

$$\Rightarrow a^2 - 9 < 0 \Rightarrow a^2 < 9 \Rightarrow -3 < a < 3 \Rightarrow a \in (-3, 3)$$

19. (b) Statement 2 is $\sqrt{n(n+1)} < n+1, n \geq 2$

$$\Rightarrow \sqrt{n} < \sqrt{n+1}, n \geq 2 \text{ which is true}$$

$$\Rightarrow \sqrt{2} < \sqrt{3} < \sqrt{4} < \sqrt{5} < \dots < \sqrt{n}$$

$$\text{Now } \sqrt{2} < \sqrt{n} \Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{\sqrt{n}}$$

$$\sqrt{3} < \sqrt{n} \Rightarrow \frac{1}{\sqrt{3}} > \frac{1}{\sqrt{n}}; \quad \sqrt{n} \leq \sqrt{n} \Rightarrow \frac{1}{\sqrt{n}} \geq \frac{1}{\sqrt{n}}$$

$$\text{Also } \frac{1}{\sqrt{1}} > \frac{1}{\sqrt{n}} \therefore \text{Adding all, we get}$$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \frac{n}{\sqrt{n}} = \sqrt{n}$$

Hence both the statements are correct and statement 2 is a correct explanation of statement-1.

20. (d) Let the roots of equation $x^2 - 6x + a = 0$ be α and 4β and that of the equation

$$x^2 - cx + 6 = 0 \text{ be } \alpha \text{ and } 3\beta. \text{ Then}$$

$$\alpha + 4\beta = 6; \quad 4\alpha\beta = a$$

$$\text{and } \alpha + 3\beta = c; \quad 3\alpha\beta = 6$$

$$\Rightarrow a = 8$$

$$\therefore \text{The equation becomes } x^2 - 6x + 8 = 0$$

$$\Rightarrow (x-2)(x-4) = 0 \Rightarrow \text{roots are 2 and 4}$$

$$\Rightarrow \alpha = 2, \beta = 1 \quad \therefore \text{Common root is 2.}$$

21. (b) Given that roots of the equation

$$bx^2 + cx + a = 0 \text{ are imaginary}$$

$$\therefore c^2 - 4ab < 0 \quad \dots(i)$$

$$\text{Let } y = 3b^2x^2 + 6bcx + 2c^2$$

$$\Rightarrow 3b^2x^2 + 6bcx + 2c^2 - y = 0$$

$$\text{As } x \text{ is real, } D \geq 0$$

$$\Rightarrow 36b^2c^2 - 12b^2(2c^2 - y) \geq 0$$

$$\Rightarrow 12b^2(3c^2 - 2c^2 + y) \geq 0$$

$$\Rightarrow c^2 + y \geq 0 \Rightarrow y \geq -c^2$$

$$\text{But from eqn. (i), } c^2 < 4ab \text{ or } -c^2 > -4ab$$

$$\therefore \text{we get } y \geq -c^2 > -4ab$$

$$\Rightarrow y > -4ab$$

22. (a) Given that $\left| z - \frac{4}{z} \right| = 2$

$$\text{Now } |z| = \left| z - \frac{4}{z} + \frac{4}{z} \right| \leq \left| z - \frac{4}{z} \right| + \frac{4}{|z|}$$

$$\Rightarrow |z| \leq 2 + \frac{4}{|z|} \Rightarrow |z|^2 - 2|z| - 4 \leq 0$$

$$\Rightarrow \left(|z| - \frac{2 + \sqrt{20}}{2} \right) \left(|z| - \frac{2 - \sqrt{20}}{2} \right) \leq 0$$

$$\Rightarrow (|z| - (1 + \sqrt{5})) (|z| - (1 - \sqrt{5})) \leq 0$$

$$\Rightarrow (-\sqrt{5} + 1) \leq |z| \leq (\sqrt{5} + 1)$$

$$\Rightarrow |z|_{\max} = \sqrt{5} + 1$$

23. (b) $x^2 - x + 1 = 0 \Rightarrow x = \frac{1 \pm \sqrt{1-4}}{2}$

$$x = \frac{1 \pm \sqrt{3}i}{2}$$

$$\alpha = \frac{1}{2} + i\frac{\sqrt{3}}{2} = -\omega^2 \quad \beta = \frac{1}{2} - i\frac{\sqrt{3}}{2} = -\omega$$

$$\alpha^{2009} + \beta^{2009} = (-\omega^2)^{2009} + (-\omega)^{2009} = -\omega^2 - \omega = 1$$

24. (b) Given equation is $e^{\sin x} - e^{-\sin x} - 4 = 0$

Put $e^{\sin x} = t$ in the given equation, we get

$$t^2 - 4t - 1 = 0$$

$$\Rightarrow t = \frac{4 \pm \sqrt{16+4}}{2} = \frac{4 \pm \sqrt{20}}{2} = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$$

$$\Rightarrow e^{\sin x} = 2 \pm \sqrt{5} \quad (\because t = e^{\sin x})$$

$$\Rightarrow e^{\sin x} = 2 - \sqrt{5} \text{ and } e^{\sin x} = 2 + \sqrt{5}$$

$$\Rightarrow e^{\sin x} = 2 - \sqrt{5} < 0 \text{ and } \sin x = \ln(2 + \sqrt{5}) > 1$$

So rejected

So, rejected

Hence given equation has no solution.

$\therefore$ The equation has no real roots.

25. (d) $f(x) = 2x^3 + 3x + k$

$$f'(x) = 6x^2 + 3 > 0 \quad \forall x \in \mathbb{R} \quad (\because x^2 > 0)$$

$$\Rightarrow f(x) \text{ is strictly increasing function}$$

$$\Rightarrow f(x) = 0 \text{ has only one real root, so two roots are not possible.}$$

26. (b) From the given system, we have

$$\frac{k+1}{k} = \frac{8}{k+3} \neq \frac{4k}{3k-1} \quad (\because \text{System has no solution})$$

$$\Rightarrow k^2 + 4k + 3 = 8k$$

$$\Rightarrow k = 1, 3$$

$$\text{If } k = 1 \text{ then } \frac{8}{1+3} \neq \frac{4.1}{2} \text{ which is false}$$

And if $k = 3$

$$\text{then } \frac{8}{6} \neq \frac{4.3}{9-1} \text{ which is true, therefore } k = 3$$

Hence for only one value of k . System has no solution.

27. (a) Given equations are

$$x^2 + 2x + 3 = 0 \quad \dots(i)$$

$$ax^2 + bx + c = 0 \quad \dots(ii)$$

Roots of equation (i) are imaginary roots.

According to the question (ii) will also have both roots same as (i). Thus

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{3} = \lambda \text{ (say)} \Rightarrow a = \lambda, b = 2\lambda, c = 3\lambda$$

Hence, required ratio is $1 : 2 : 3$

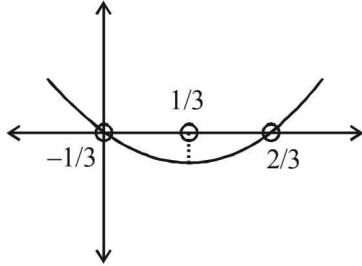
28. (c) Consider $-3(x - [x])^2 + 2[x - [x]] + a^2 = 0$

$$\Rightarrow 3\{x\}^2 - 2\{x\} - a^2 = 0 \quad (\because x - [x] = \{x\})$$

Quadratic Equation and Inequalities (Inequalities)

$$\Rightarrow 3\left(\{x\}^2 - \frac{2}{3}\{x\}\right) = a^2, a \neq 0$$

$$\Rightarrow a^2 = 3\{x\}\left(\{x\} - \frac{2}{3}\right)$$



Now, $\{x\} \in (0, 1)$ and $-\frac{2}{3} \leq a^2 < 1$ (by graph)

Since, x is not an integer

$$\therefore a \in (-1, 1) - \{0\} \Rightarrow a \in (-1, 0) \cup (0, 1)$$

29. (b) Let p, q, r are in AP

$$\Rightarrow 2q = p + r \quad \dots(i)$$

$$\text{Given } \frac{1}{\alpha} + \frac{1}{\beta} = 4 \Rightarrow \frac{\alpha + \beta}{\alpha\beta} = 4$$

$$\text{We have } \alpha + \beta = -q/p \text{ and } \alpha\beta = \frac{r}{p}$$

$$\Rightarrow \frac{-\frac{q}{p}}{\frac{r}{p}} = 4 \Rightarrow q = -4r$$

From (i), we have

$$2(-4r) = p + r \Rightarrow p = -9r$$

$$q = -4r$$

$$\text{Now } |\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

$$= \sqrt{\left(\frac{-q}{p}\right)^2 - \frac{4r}{p}} = \frac{\sqrt{q^2 - 4pr}}{|p|}$$

$$= \frac{\sqrt{16r^2 + 36r^2}}{|-9r|} = \frac{2\sqrt{13}}{9}$$

$$30. (a) \alpha, \beta = \frac{6 \pm \sqrt{36+8}}{2} = 3 \pm \sqrt{11}$$

$$\alpha = 3 + \sqrt{11}, \beta = 3 - \sqrt{11}$$

$$\therefore a_n = (3 + \sqrt{11})^n - (3 - \sqrt{11})^n$$

$$\frac{a_{10} - 2a_8}{2a_9}$$

$$= \frac{(3 + \sqrt{11})^{10} - (3 - \sqrt{11})^{10} - 2(3 + \sqrt{11})^8 + 2(3 - \sqrt{11})^8}{2[(3 + \sqrt{11})^9 - (3 - \sqrt{11})^9]}$$

$$= \frac{(3 + \sqrt{11})^8 [(3 + \sqrt{11})^2 - 2] + (3 - \sqrt{11})^8 [2 - (3 - \sqrt{11})^2]}{2[(3 + \sqrt{11})^9 - (3 - \sqrt{11})^9]}$$

$$= \frac{(3 + \sqrt{11})^8 (9 + 11 + 6\sqrt{11} - 2) + (3 - \sqrt{11})^8 (2 - 9 - 11 + 6\sqrt{11})}{2[(3 + \sqrt{11})^9 - (3 - \sqrt{11})^9]}$$

$$= \frac{6(3 + \sqrt{11})^9 - 6(3 - \sqrt{11})^9}{2[(3 + \sqrt{11})^9 - (3 - \sqrt{11})^9]} = \frac{6}{2} = 3$$

$$\dots(ii) \quad 31. (c) (x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$$

Case I

$x^2 - 5x + 5 = 1$ and $x^2 + 4x - 60$ can be any real number

$$\Rightarrow x = 1, 4$$

Case II

$x^2 - 5x + 5 = -1$ and $x^2 + 4x - 60$ has to be an even number

$$\Rightarrow x = 2, 3$$

where 3 is rejected because for $x = 3$, $x^2 + 4x - 60$ is odd.

Case III

$x^2 - 5x + 5$ can be any real number and $x^2 + 4x - 60 = 0$

$$\Rightarrow x = -10, 6$$

$$\Rightarrow \text{Sum of all values of } x = -10 + 6 + 2 + 1 + 4 = 3$$